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Does Unemployment Cause Future Unemployment? Definitions, Questions and Answers from a Continuous Time Model of Heterogeneity and State Dependence

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INTRODUCTION

Recent research demonstrates that, the greater the number of previous spells of unemployment and the longer their duration, the more likely is the event that an individual will be unemployed at a point in time. Two explanations have been advanced to interpret this finding. The first is rooted in economic theory; the second is based solely on statistical considerations.

Under the first explanation, past unemployment (including previous time spent in a current unemployment spell) alters preferences, prices or constraints that determine, in part, future unemployment. For example, unemployment may lead to a loss of work experience, which will alter future prospects of employability. As another example, if workers are heterogeneous in unobserved components of ability, firms may use unemployment records in their hiring decisions if the knowledge that a worker has been unemployed is useful in sorting out the worker's position in the population distribution of heterogeneity and if firms place sufficient value on such information in making worker hiring and investment decisions. In both cases, prior unemployment experience has a genuine behavioural effect in the sense that an otherwise identical individual who did not experience unemployment would behave differently in the future than an individual who experienced unemployment. Structural relationships of this sort give rise to true state dependence as defined in this paper.

A second explanation for the empirical regularity is that individuals differ in certain unmeasured variables that influence their probability of experiencing unemployment but that are not influenced by the experience of unemployment. If these unmeasured variables are correlated over time and are not properly controlled, previous unemployment may appear to be a determinant of future unemployment solely because it is a proxy variable for temporally correlated unobservables. Improper treatment of unmeasured variables gives rise to a conditional relationship between future and past unemployment due solely to uncontrolled heterogeneity.

This paper presents statistical methods for testing for the presence of true state dependence. Our methods are based on extensions of existing models for continuous-time discrete-state Markov processes. In this paper we develop a new model with general forms of state dependence. With this model it is possible to isolate four main types of structural state dependence.

The first type of state dependence is Markovian. In a short interval of time, the probability that an employed worker will become unemployed differs from the probability that an unemployed worker will remain unemployed. Such state dependence may arise for many reasons, transaction costs being a prominent one. The second type of state dependence is termed "occurrence dependence". The

number of previous spells of unemployment affects the probability that a worker will become or remain unemployed. Such state dependence may arise if employers use unemployment records in their hiring and firing decisions. The third type of state dependence is “duration dependence”. The probability of remaining unemployed depends on the length of time the worker has been unemployed in his current unemployment spell. Such state dependence may arise as a consequence of declining assets during the unemployment spell or because horizons are shortened during the unemployment spell. The fourth type of state dependence is termed “lagged duration dependence”. The probabilities of remaining unemployed or becoming unemployed depend on the *lengths* of previous unemployment spells. Such state dependence can arise if unemployment results in a loss of productivity-enhancing work experience.

Tests for Markov dependence against independence in a continuous-time model raise certain technical problems which are not discussed here. Moreover, it is noted below that in a continuous-time model Markov dependence is a natural starting point. Tests for the other three types of state dependence occupy our attention in this paper.

A major conclusion of this paper is that the conditions that must be satisfied for identification of each of the three types of state dependence are different. Occurrence dependence is the least demanding. Duration dependence is the most demanding and requires strong prior information that is usually not at our disposal. Identification conditions for lagged duration dependence are analogous to identification conditions in econometric models with lagged values of the dependent variable and serially correlated errors.

Three approaches to testing for structural state dependence are developed in our work. The first approach draws on and extends procedures developed in the literature on accident-proneness (Bates and Neyman, 1952; Arbous and Kerrich, 1951). The second approach draws on and extends regression models for point process data due to Cox and Lewis (1966). The third approach is based on maximum likelihood methods and is developed more extensively in other work (Heckman and Borjas, 1979; Flinn and Heckman, 1980).

There are three main advantages of the methods discussed in this paper over previous work based on discrete-time models (Heckman, 1977). The first is that the underlying probability process is developed in continuous time so that well-known time-aggregation problems that plague discrete-time models can be avoided. With continuous-time models, it is not necessary to select an arbitrary time unit in order to conduct tests for true state dependence (on the topic of time aggregation see Singer and Spilerman, 1976; Singer, 1978; Bergstrom, 1976).

The second advantage of our methods is that they can be linked to existing theoretical models of labour market turnover developed by Holt (1970) and Burdett and Mortenson (1977).

The main advantage of our models is that they provide a framework for more precise definitions of the vague concept of state dependence that has come into wide use in the recent empirical literature in labour economics. Our definitions help to clarify the nature of true state dependence.

This paper is divided into three parts. In the first section, we review some familiar facts about continuous-time discrete-state Markov models. We then generalize these models to accommodate the three additional types of state

dependence discussed above. Having set up the basic model, we then introduce heterogeneity into it. In the second section, we develop procedures for estimating the models and testing for true state dependence. In the third section, we present some empirical tests for true state dependence based on a sample of 122 young men who graduated from high school in 1969 and were interviewed for 30 consecutive months after graduation. These results are very tentative but they illustrate the approach suggested in this paper. In our data, we find little evidence that current unemployment causes future unemployment.

I. A TWO-STATE MODEL OF EMPLOYMENT AND UNEMPLOYMENT

(a) *A continuous-time discrete-state Markov model of employment and unemployment*

In order to focus the discussion and motivate the empirical work discussed below, we consider a model of labour market turnover that appears in work by Holt (1970) and Burdett and Mortenson (1977). To simplify the discussion, workers are assumed to be either employed or unemployed so that no one drops out of the labour force. We further suppose that there is no distinction between unemployment on recall and permanent layoff unemployment. We initially ignore heterogeneity in measured and unmeasured exogenous variables. None of these assumptions is essential and most are relaxed in later sections of this paper or in other work (Heckman and Borjas, 1979; Flinn and Heckman, 1980). The simple model presented in this section enables us quickly to survey a rich literature, largely ignored in economics, that forms the background for our work.

A worker can be in one of two states at time t : the employment state or the unemployment state. $P_1(t)$ is the probability that a worker is employed at time t ; $P_2(t)$ is the probability that he is unemployed. Suppose that in a "small" interval of time, Δt , the probability that an individual changes state more than once is negligible. This "Poisson-like" assumption is a key feature of the model.

Consider a worker employed at time t . The probability that in an interval t to $t + \Delta t$ the worker will become unemployed (i.e., move from state one to state two) is defined to be $a_{12}\Delta t + o(\Delta t)$ where $o(\Delta t)$ is a term with the property

$$\lim_{\Delta t \rightarrow 0} \frac{o(\Delta t)}{\Delta t} \rightarrow 0.$$

The probability that the worker will continue on in employment is defined to be $(a_{11}\Delta t + 1) + o(\Delta t)$. For a worker unemployed at time t similar terms can be defined. The probability that in an interval t to $t + \Delta t$ the worker will become employed is $a_{21}\Delta t + o(\Delta t)$, while the probability that the worker will continue in unemployment is $(a_{22}\Delta t + 1) + o(\Delta t)$. From the definitions, it is clear that a_{12} , $a_{21} > 0$ while a_{22} and $a_{11} < 0$.

Behavioural content can be given to the model (see especially the discussion in Burdett and Mortenson, 1977). The instantaneous transition probabilities a_{ij} are determined by firm layoff policies and worker characteristics. In the general model developed in this paper, the a_{ij} can be parameterized to depend on both measured and unmeasured worker, firm, and labour market characteristics, and the entire labour market history of the worker. As the model stands, it is Markovian so that the transitions depend solely on the current state in which the worker is located.

Before the Markov assumption is relaxed, it is useful to collect and summarize certain well-known results about the Markov model. The probability that a worker is employed at time $t + \Delta t$ is

$$P_1(t + \Delta t) = (a_{11}\Delta t + 1)P_1(t) + (a_{21}\Delta t)P_2(t) + o(\Delta t).$$

A worker is employed at time $t + \Delta t$ either because he is employed at time t (this occurs with probability $P_1(t)$) and continues on in employment (this occurs with probability $(a_{11}\Delta t + 1) + o(\Delta t)$, or because he is unemployed at time t (this occurs with probability $P_2(t)$) and becomes employed (this occurs with probability $a_{21}\Delta t + o(\Delta t)$). The change in the probability of employment over the interval $t, t + \Delta t$ is

$$P_1(t + \Delta t) - P_1(t) = (a_{11}\Delta t)P_1(t) + (a_{21}\Delta t)P_2(t) + o(\Delta t).$$

Dividing through by Δt , and passing to the limit $\Delta t \rightarrow 0$, we reach

$$(1) \quad \dot{P}_1(t) = a_{11}P_1(t) + a_{21}P_2(t).$$

By a parallel argument,

$$(2) \quad \dot{P}_2(t) = a_{12}P_1(t) + a_{22}P_2(t)$$

since $P_1(t) + P_2(t) = 1$, $\dot{P}_1(t) + \dot{P}_2(t) = 0$ so that $a_{11} + a_{12} = 0$ and $a_{21} + a_{22} = 0$.¹

It is well known (see, e.g., Cox and Miller, 1965; Tuma *et al.*, 1979) that this model can be interpreted as an alternating Poisson process with exponential “waiting time”. Consider a worker employed at time t . At each instant he is subject to the “risk” that he becomes unemployed. The instantaneous rate of entering unemployment from employment is a_{12} ($= -a_{11}$). The distribution of time to unemployment for a currently employed worker (which is also the distribution of time in the state for a worker just entering the state) is an exponential function that is *independent* of the time the worker becomes employed, and that is independent of the time period in which the event occurs. Both of these assumptions are relaxed below.

The distribution function of time to unemployment (i.e., the length of time employed of a currently working individual, t_{12}), F_{12} , is

$$(3) \quad F_{12}(t_{12}) = 1 - \exp(-a_{12}t_{12})$$

so that the density is

$$f_{12}(t_{12}) = a_{12} \exp(-a_{12}t_{12}).$$

The mean length of time employed in a spell of employment is given by

$$E(t_{12}) = \frac{1}{a_{12}}.$$

Given the Markovian assumption, and assuming a stationary environment (so that a_{12} does not depend on time varying variables), $1/a_{12}$ is the mean duration of remaining time to unemployment for *any* employed worker, irrespective of how long he has been employed. In the language of renewal theory there is no duration dependence in the distribution of time to unemployment. Using standard methods, we relax this assumption below.

By a parallel argument, the distribution function (F_{21}) of time to employment of a currently unemployed individual (t_{21}) is

$$(4) \quad F_{21}(t_{21}) = 1 - \exp(-a_{21}t_{21})$$

so that the density is

$$f_{21}(t_{21}) = a_{21} \exp(-a_{21}t_{21})$$

and the mean length of stay in the unemployment state for a worker who just becomes unemployed is

$$E(t_{21}) = \frac{1}{a_{21}}.$$

Two renewal processes govern the labour market behaviour of workers. The first process is an unemployment process. The second is an employment process. The first operates when the individual is employed and the second operates when the individual is unemployed. Figure 1 helps to illustrate the model. Consider a worker who starts life employed. At time t_1 , he becomes unemployed, at time t_2 he becomes employed, and so forth. The distribution of time to the first spell of unemployment is given by equation (3). This is also the distribution of the length of the first interval of employment. The distribution of the length of the first interval of unemployment ($t_2 - t_1$) is given by equation (4). In the simple model considered here, the worker can be in only one of two states, and given his initial state the distribution of duration times in each state fully characterizes his labour market history.

The crucial assumptions that underlie this simple process (apart from the assumption that the probability of two or more events in a small interval is negligible) are (a) that the length of time in a state in the current spell does not affect the rate of transition out of the state; and (b) that the transition functions do not depend on the labour market history from preceding spells. Using terms to be defined more precisely below, the first assumption rules out duration dependence while the second assumption rules out occurrence dependence and lagged duration dependence.

(b) Duration, occurrence and lagged duration dependence

The model just discussed is well known. Versions of it have been used in sociology for some time (see, e.g., Coleman, 1964; Tuma *et al.*, 1979; Singer and Spilerman, 1976). Explanatory variables have been introduced into the model (to

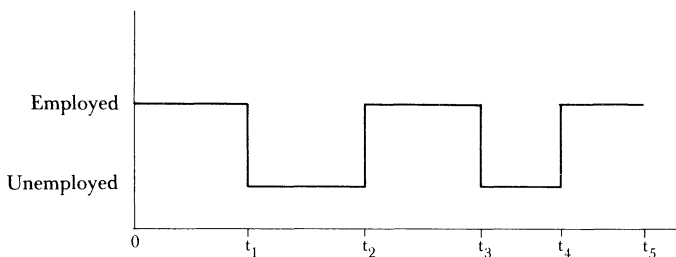


FIGURE 1

parameterize a_{ij}), and unobservable characteristics that determine the a_{ij} have also been introduced into such models.² (See Singer and Spilerman, 1976; Tuma, 1979.) In this section, we alter the basic structure of the simple Markov model to generate a new model that captures the essential features of the four kinds of structural state dependence discussed in the Introduction.

The continuous-time discrete-state Markov model admits a special kind of state dependence. It allows for the possibility that an individual who is currently unemployed has a different probability of being unemployed in the next instant $(1 + (a_{22})\Delta t)$ than an individual who is currently employed $(a_{12}\Delta t)$. Moreover, this model captures the intuitively appealing notion that, as time intervals become small, the probability of transition from a given state becomes negligible, i.e.,

$$\lim_{\Delta t \rightarrow 0} a_{12}\Delta t \rightarrow 0, \quad \lim_{\Delta t \rightarrow 0} a_{21}\Delta t \rightarrow 0$$

so that the probability of staying on in a state approaches unity:

$$\lim_{\Delta t \rightarrow 0} (1 + a_{11}\Delta t) \rightarrow 1, \quad \lim_{\Delta t \rightarrow 0} (1 + a_{22}\Delta t) \rightarrow 1.^3$$

Economic theory suggests more general types of state dependence than the sort permitted in the simple Markov model. One reason for such dependence, popularly known as the “scar theory” of unemployment, postulates that workers who have been unemployed are altered by their experience of unemployment. Schweitzer and Smith (1974) develop a model in which the experience of unemployment, or its duration, or both, cause workers to drop out of the labour force. One can also argue, as has Phelps (1972), that unemployment experience alters one’s future probability of being unemployed because individuals lose valuable work experience while they are unemployed, or because they are marked as “losers” by potential employers.

We initially consider the case that the *event* of unemployment alters the future probability of becoming unemployed. For the moment, we put to the side the possibility that the *duration* of unemployment spells affects the probability of being unemployed.

The most direct way to introduce such dependence into the framework developed above is to modify the “waiting time” to unemployment distribution (equation (3)). Let $a_{12}^{(l)}$ be the transition rate to unemployment of a currently employed worker who has experienced l spells of unemployment. If $a_{12}^{(l)} \neq a_{12}^{(l')}$, $l \neq l'$, the risk of transiting to unemployment depends on the previous spells of unemployment. For example, if $a_{12}^{(l)} > a_{12}^{(l-1)}$, the mean duration in the employed state for a worker with l spells of unemployment ($1/a_{12}^{(l)}$) is less than the mean duration in the state of *employment* for an individual with $l-1$ spells ($1/a_{12}^{(l-1)}$). An alternative way of stating the same point is that the distribution of waiting time to unemployment for those workers who have just become unemployed has more probability mass near the origin for those with l spells of unemployment than for those with $l-1$ spells. This may be displayed graphically in Figures 2 and 3, which assume $a_{12}^{(l)} > a_{12}^{(l-1)}$.

This way of defining “state dependence” is very similar to the way “true contagion” is defined in the literature on accident-proneness (see, e.g., Feller,

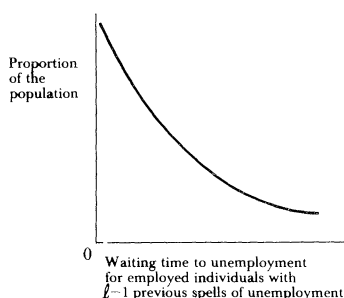


FIGURE 2

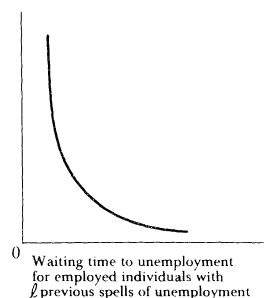


FIGURE 3

1943; Bates and Neyman, 1952; Arbous and Kerrich, 1951). In that literature, experience of an accident is said to *cause* future accidents if the mean duration time to the next accident is lower for individuals with a greater accident record.⁴

This concept of state dependence can be generalized in two ways. First, a_{21} , the instantaneous probability of transiting to employment from a current state of unemployment, can be indexed by the number of spells of employment up to the current spell $a_{21}^{(k)}$, and the definition of state dependence can be broadened to include the possibility that $a_{21}^{(k)} \neq a_{21}^{(k')}$, $k \neq k'$. Second, both episodes of employment and unemployment can determine the a_{ij} so that in general the transition rate from the m th state to the n th state, $a_{m,n}$, depends on l (the number of previous spells of unemployment), k (the number of previous spells of employment), or both (thus we introduce $a_{m,n}^{(l,k)}$). Clearly, however, in a two-state model the number of spells of unemployment and the number of spells of employment are not independent, and differ by at most one.⁵

The concept of state dependence just discussed extends the definition of state dependence in the continuous-time Markov model to account for the effect of previous *occurrence* of employment and unemployment on the instantaneous rates of transition out of the employment and unemployment states. This form of state dependence is termed “occurrence dependence”. Within episodes of employment or unemployment, the distributions of exit times from a state are assumed to be exponential, as in the model of sub-section (a) above. Unlike the situation in that model, the parameter of each exit time distribution depends on the *number* of previous episodes of employment or unemployment. Further properties of this model are developed below, but before these are considered it is useful to explore another modification of the discrete-state continuous-time Markov model. This is a model with *duration dependence*.

In this model, occurrence dependence is ignored. However, the distribution of exit times is assumed to be non-exponential (see Cox, 1962). It is well known (see Feller, 1968) that the exponential exit time distribution is the only exit time distribution that does not possess a “memory” in the sense that the conditional distribution of exit time from a state given previous time in the state is the same for all values of time previously spent in the state.

In place of equation (3) for the simple Markov model, the duration dependent or semi-Markov process has a general distribution function $F_{12}(t_{12})$ with density $f_{12}(t_{12})$. As in the model of sub-section (a), this distribution does *not* depend on

previous spells or previous durations. Unlike the model in that section, the exit time distribution is non-exponential.

To simplify our discussion of duration dependence it is helpful to introduce the concept of a hazard function, which is defined as the conditional density of exit time from a state given the amount of time spent in the state in the current spell. The hazard rate for exit time t_{12} , given that the time spent in the employment state in a current spell of unemployment is at least t_{12} , is defined as

$$h_{12}(t_{12}) = \frac{f_{12}(t_{12})}{1 - F_{12}(t_{12})}.$$

The hazard rate for exit time t_{21} , given that time spent in the unemployment state in a current spell of unemployment is t_{21} , is defined as

$$h_{21}(t_{21}) = \frac{f_{21}(t_{21})}{1 - F_{21}(t_{21})}.$$

For the continuous-time Markov model, $h_{21}(t_{21}) = a_{21}$ and $h_{12}(t_{12}) = a_{12}$.

Given the hazard functions, one can derive the distribution functions. Thus

$$(5) \quad F_{ij}(t_{ij}) = 1 - \exp \left\{ - \int_0^{t_{ij}} h_{ij}(u) du \right\} \\ f_{ij}(t_{ij}) = h_{ij}(t_{ij}) \exp \left\{ - \int_0^{t_{ij}} h_{ij}(u) du \right\}, \quad i, j = 1, 2, i \neq j.^6$$

Duration dependence is defined to exist if the hazard function depends on the length of time elapsed in the state in the current spell. Positive duration dependence is said to exist if

$$\frac{\partial h_{ij}(t_{ij})}{\partial t_{ij}} > 0, \quad i, j = 1, 2, i \neq j$$

while negative duration dependence is said to exist if the inequality is reversed. Negative duration dependence implies that the longer an individual has been in the state the less likely he is to leave the state. A model with negative duration dependence captures the basic idea of "firm-specific human capital" applied to labour turnover. The longer a worker has been at a firm, the greater the amount of specific investment invested in him, and hence the less likely he is to leave the firm (Tuma, 1976). In an analysis of the duration of unemployment spells based on the theory of search, the hypothesis of a declining reservation wage owing to cash constraints and a diminished horizon translates into positive duration dependence.

A complete catalogue of non-exponential distributions used to parameterize duration dependence is available in Cox (1962, pp. 15–24). For a general parametric family see Farewell and Prentice (1977). For example, the Weibull distribution sets

$$(6) \quad h_{ij}(t_{ij}) = a_{ij} \gamma_{ij} (t_{ij})^{\gamma_{ij}-1}, \quad i, j = 1, 2, i \neq j.$$

If $\gamma_{ij} = 1$, there is no duration dependence since the distribution is exponential. If $\gamma_{ij} > 1$, there is positive duration dependence, and if $\gamma_{ij} < 1$, there is negative duration dependence. It is helpful to note that if time is transformed to Weibull time ($\tau_{ij} = (t_{ij})^{\gamma_{ij}}$), the distribution of t_{ij} is exponential in Weibull time. On this transformed time scale, a_{ij} is a transition intensity.

Another distribution of exit times that is used below is the log-normal distribution, which has hazard rate

$$(7) \quad h_{ij}(t_{ij}) = \frac{(2\pi\sigma_{ij})^{-1/2} \exp\left[-\frac{(\ln(\rho_{ij}t_{ij}))^2}{2\sigma_{ij}}\right] \frac{1}{t_{ij}}}{1 - \int_0^{t_{ij}} (2\pi\sigma_{ij})^{-1/2} \exp\left[-\frac{(\ln(\rho_{ij}u))^2}{2\sigma_{ij}}\right] \frac{du}{u}}.$$

$$i, j = 1, 2, i \neq j.$$

The hazard function rises and then falls with t_{ij} (see Cox, 1962). “ $\ln t_{ij}$ ” is normally distributed with mean $(-\ln \rho_{ij})$ and variance σ_{ij} .

Piecing together the two waiting-time distributions for exit to employment and exit to unemployment generalizes the continuous-time Markov model to allow for duration dependence for both employment and unemployment. Let $t_{ij}^{(l)}$ be the duration of the l th spell of event i . The density of duration times for a worker who begins life employed, who has had $J - 1$ spells of unemployment, and who has been working α time units in his J th uncompleted employment spell is

$$(8) \quad \prod_{j=1}^{J-1} f_{21}(t_{21}^{(j)}) \prod_{j=1}^{J-1} f_{12}(t_{12}^{(j)}) \{1 - F_{12}(\alpha)\}.$$

Duration dependence is the effect of current *duration* in a state on the probability of exiting the state. Occurrence dependence is the effect of previous *occurrences* of events on the probability of exiting a state. A natural extension of these ideas considers the effect of durations of previous spells on the probability of exiting a state. This type of state dependence, termed “lagged duration dependence” in this paper, captures the notion implicit in Phelps (1972) that the *lengths* of spells of unemployment and employment affect subsequent unemployment and employment waiting-time distributions. Singer and Spilerman (1976, pp. 460–461) have considered a special version of such models.

A general representation of lagged duration dependence writes the hazard rate for the l th spell of event i as

$$(9) \quad h_{ij}(t_{ij}^{(l)}) = g_{ij}(t_{ij}^{(l)}, t_{ij}^{(l-1)}, \dots, t_{ji}^{(l)}, \dots, t_{ji}^{(1)}) \quad i, j = 1, 2, i \neq j.$$

Note that $t_{ji}^{(l)}$ is entered as an argument in the hazard function. If the individual begins life in state j this is appropriate. If the individual begins life in state i , we establish the convention that the partial derivative of the hazard function with respect to $t_{ji}^{(l)}$ is always zero.

A simple parametric representation of lagged duration dependence in the Weibull framework is

$$(10a) \quad h_{ij}(t_{ij}^{(l)}) = a_{ij}(t_{ij}^{(l)})^{\nu_{ij}-1} \gamma_{ij} \exp - \left\{ \sum_{k=1}^{l-1} \eta_{1i}(k) \ln t_{ij}^{(l-k)} + \sum_{k=0}^{l-1} \eta_{2i}(k) \ln t_{ji}^{(l-k)} \right\}$$

$$i, j = 1, 2, i \neq j$$

where the $\eta_{1i}(s)$, $\eta_{2i}(p)$, $p = 1, \dots, l-1$, $s = 0, \dots, l-1$ are parameters. If $\eta_{mi}(p) = 0$ for all p , durations of previous spells of event m do not affect the distribution of the waiting time for the l th spell of event i . A simple parametric

representation of lagged duration dependence in the log-normal model writes ρ_{ij} in equation (7) as

$$(10b) \quad \rho_{ij} = \tilde{a}_{ij} \exp - \left\{ \sum_{k=1}^{l-1} \tilde{\eta}_{1l}(k) \ln t_{ij}^{(l-1)} + \sum_{k=0}^{l-1} \tilde{\eta}_{2l}(k) \ln t_{ji}^{(l-k)} \right\}, \quad i \neq j$$

where the tilde is used to distinguish log-normal from Weibull parameters.

A general model that combines duration dependence, occurrence dependence and lagged duration dependence writes the hazard function as

$$h_{ij}^{(l)}(t_{ij}^{(l)}) = g_{ij}^{(l)}(t_{ij}^{(l)}, \dots, t_{ij}^{(1)}, t_{ji}^{(l)}, \dots, t_{ji}^{(1)}), \quad i, j = 1, 2, i \neq j.$$

If the first argument of the function has a zero derivative for all values of the right-hand side variables, there is no duration dependence. If the remaining arguments have zero derivatives for all values of the right-hand side variables there is no lagged duration dependence. If the function g is stationary across spells ($l = 1, \dots$), there is no occurrence dependence.

In constructing these definitions we have abstracted from the problem that the time at which events occur may affect subsequent effects of employment and unemployment. Thus a spell of unemployment suffered in the Great Depression will convey less information about a worker than a spell of unemployment suffered in a tight labour market. To capture this effect, one can introduce interactions between past events and calendar time (or other variables that are indexed by calendar time) at which past events occur. A similar effect may arise as a function of age. Thus, if younger workers are known to be “unemployment-prone”, knowledge that a young worker has been unemployed conveys less information. This reasoning suggests incorporating age interactions as well.

A parametric representation of the general model in a Weibull framework writes

$$(11a) \quad h_{ij}^{(l)}(t_{ij}^{(l)}) = \gamma_{ij}^{(l)} a_{ij}^{(l)} (t_{ij}^{(l)})^{\gamma_{ij}^{(l)}-1} \exp - \left\{ \sum_{j=1}^{l-1} \eta_{1l}^{(l)}(k) \ln t_{ij}^{(l-k)} + \sum_{k=0}^{l-1} \eta_{2l}^{(l)}(k) \ln t_{ji}^{(l-k)} \right\},$$

$$i \neq j$$

where the superscripts on the parameters denote the l th spell of event i . A parametric representation of the general model for the log-normal case substitutes for ρ_{ij} in equation (7),

$$(11b) \quad \rho_{ij}^{(l)} = \tilde{a}_{ij}^{(l)} \exp - \left\{ \sum_{k=1}^{l-1} \tilde{\eta}_{1l}^{(l)}(k) \ln t_{ij}^{(l-k)} + \sum_{k=0}^{l-1} \tilde{\eta}_{2l}^{(l)}(k) \ln t_{ji}^{(l-k)} \right\}, \quad i \neq j.$$

(c) *Heterogeneity*

This section extends the preceding models to allow for heterogeneity. Heterogeneity is defined as unmeasured and measured variables that differ among individuals and that may differ over time for the same individual although the term is usually reserved for unobservables as perceived by the data analyst. Here we consider both types of heterogeneity.

It is useful to distinguish between pure heterogeneity and state dependent heterogeneity. The essential feature of pure heterogeneity is that the joint distribution of pure heterogeneity components is not determined by outcomes of

the employment and unemployment processes. Examples of pure heterogeneity include individual characteristics which, while they may change over time, do not change as a function of outcomes of the employment and unemployment processes. Pure heterogeneity components are exogenous to the process. In the language of Sims (1977), pure heterogeneity components cause but are not caused by outcomes of the unemployment and employment processes. Thus the conditional distribution of pure heterogeneity components given all lagged values of such components and given all lagged values of outcomes of the employment and unemployment processes is the same as the conditional distribution of pure heterogeneity components given all previous pure heterogeneity components. Pure heterogeneity in unobserved components is commonly termed heterogeneity in the literature.

The unobservables may include state dependent components as well. For example, a spell of unemployment may inflict different economic losses on different people, and one may not be able to fully measure the differences in the losses. The unobserved loss components are a direct outcome of the process itself and so are not exogenous to the process. Such unobserved state dependence components may affect future unemployment and employment and are affected by past unemployment and employment. The joint distribution of state dependent heterogeneity components is determined, in part, by the history of the process unlike the joint distribution of pure heterogeneity components. The procedures developed below allow for both types of unobserved heterogeneity. It is possible to find no state dependence in parameters associated with observed variables while finding state dependence in the parameters of the distribution of unobservables. In any event, improper treatment of heterogeneity in dynamic models can lead to seriously biased estimates of state dependence effects.⁷

Any empirically useful treatment of heterogeneity requires some parametric structure. A fully general analysis that makes exit time distributions arbitrary functions of measured and unmeasured components is vacuous. The strategy adopted in this paper is to adopt simple parameterizations of the hazard function. For the present purposes, we restrict our analysis to the Weibull family and log-normal family, although we indicate how more general approaches may be pursued.

It is convenient to commence our analysis under the assumption that each spell of an event has a set of spell-specific explanatory variables that remain constant within each spell but may change across spells. For many problems this assumption is artificial, but for other problems it is less so. For example, in an analysis of a worker's tenure at a firm, characteristics of the worker at entry may play the dominant role.

Let $\mathbf{Z}_{ij}^{(l)}$ denote a vector of exogenous explanatory variables for the l th spell of event i , and let $\mathbf{V}_{ij}^{(l)}$ be unmeasured explanatory variables for the spell.⁸ We suppress subscripts for individual observations to simplify the notation. The unmeasured components are freely correlated across spells and events. Since transition intensities are required to be positive, it is convenient to adopt the following parameterizations for the Weibull model which has clear antecedents in work by Cox and Lewis (1966) and Tuma *et al.* (1979):

$$(12a) \quad a_{ij}^{(l)} = \exp - (\beta_{ij}^{(l)} \mathbf{Z}_{ij}^{(l)} + \mathbf{V}_{ij}^{(l)})$$

where $\beta_{ij}^{(l)}$ is a suitably dimensioned vector of parameters. A corresponding

parameterization for the log-normal model writes

$$(12b) \quad \rho_{ij}^{(l)} = \exp - (\tilde{\beta}_{ij}^{(l)} \mathbf{Z}_{ij}^{(l)} + \tilde{\mathbf{V}}_{ij}^{(l)}).$$

The $\mathbf{Z}_{ij}^{(l)}$ vector can be augmented to include lagged durations.

The most conventional specification of unmeasured pure heterogeneity assumes a permanent component model so that $\mathbf{V}_{ij}^{(l)} = \phi$ for all l and $i, j = 1, 2, i \neq j$. This model captures the notion that individual characteristics are stable over time but fails to capture the notion that the sign of the effect of the unmeasured variables on the a_{ij} parameters may be different in the employed state than in the unemployed state. For example, if ϕ is ability, it is plausible that ϕ may have a negative effect on transitions out of employment and a positive effect on transitions out of unemployment.

Another extreme specification—that the $\mathbf{V}_{ij}^{(l)}$ are mutually independent for all l, i and j ($i \neq j$)—rules out persistence of unmeasured characteristics across spells. Much previous empirical work in labour economics contradicts such an assumption, so that this specification is not very interesting.

A more flexible model is a one-factor scheme that structures the disturbance term in the following way for the Weibull and log-normal models

$$\mathbf{V}_{ij}^{(l)} = \mathbf{C}_{ij}^{(l)} \phi, \quad \tilde{\mathbf{V}}_{ij}^{(l)} = \tilde{\mathbf{C}}_{ij}^{(l)} \phi \quad i, j = 1, 2, i \neq j$$

where $\mathbf{C}_{ij}^{(l)}$ and $\tilde{\mathbf{C}}_{ij}^{(l)}$ are factor-loading coefficients. If all the factor-loading coefficients are identical, the permanent component model is generated. Interpreting ϕ as an omitted person-specific effect that arises from pure heterogeneity, allowing the factor loadings to differ across consecutive spells of an event builds in the possibility of state dependence in the distribution of unobservables.

A more complete discussion of identification in this model is presented elsewhere (Heckman and Borjas, 1979). When regression approaches are discussed below, we return to this topic. We note for the moment that the one-factor scheme is easily extended to allow for spell-specific innovations in unobserved components that are independent across spells. In a sense, such a specification is redundant. Given ϕ , the choice of an exit time distribution generates a spell-specific independently distributed component (see equations 20(a) and 20(b) below).

The restriction that explanatory variables do not change within a spell is unappealing in many applications. Full details of a procedure that relaxes this assumption are presented elsewhere (see Flinn and Heckman, 1980). Here we simply indicate the procedure.

Write the hazard function for the l th spell of event i which commences at calendar time τ as

$$h_{ij}^{(l)}(t_{ij}^{(l)}) = h_{ij}^{(l)}[t_{ij}^{(l)}, \mathbf{Z}_{ij}^{(l)}(t_{ij}^{(l)} + \tau), t_{ij}^{(l-1)}, \dots, t_{ij}^{(1)}, t_{ji}^{(l)}, \dots, t_{ji}^{(1)}, \mathbf{V}_{ij}^{(l)}(t_{ij} + \tau)].$$

$\mathbf{Z}_{ij}^{(l)}(t_{ij}^{(l)} + \tau)$ is the value at time $t_{ij}^{(l)} + \tau$ of the exogenous variables relevant for determining transitions from the l th spell of event i to event j . $\mathbf{V}_{ij}^{(l)}(t_{ij}^{(l)} + \tau)$ is the value of the disturbance component at time $t_{ij}^{(l)} + \tau$. The density function for $t_{ij}^{(l)}$, given $\mathbf{V}_{ij}^{(l)}(u + \tau)$ for all values $u < t_{ij}^{(l)}$, is

$$(13) \quad h_{ij}^{(l)}(t_{ij}^{(l)}) \exp \left(- \int_0^{t_{ij}^{(l)}} h_{ij}^{(l)}(u) du \right).$$

To adapt this density for empirical work, one breaks calendar time into discrete time periods within which the heterogeneity components are constant. The integral

inside the exponential is replaced by a finite sum. Given such densities, it is straightforward to write down the likelihood conditional on the unobserved components. These components can be integrated out or treated as fixed effects and estimated along with the other parameters of the model.

In our companion paper we demonstrate that maximum likelihood procedures that ignore heterogeneity when it is present result in downward-biased estimates of duration dependence effects (so that negative duration dependence or the finding of the existence of “specific human capital” tends to be generated when there is no duration dependence). This result is true in any sample, and is not just an asymptotic result. Other examples of bias arising from uncorrected heterogeneity are presented in that paper. (See also Flinn and Heckman, 1980.)

(d) Heterogeneity and state dependence: formal definitions and empirical difficulties

Before turning to the empirical procedures presented in the next section, it is useful to take stock of the new concepts of state dependence introduced in this section and to present general definitions of occurrence, duration, and lagged duration dependence that do not depend on specific functional forms for exit time distributions and heterogeneity. We do not discuss Markov dependence, which is assumed to be present. The message of this sub-section is that, without strong distributional assumptions about the nature of heterogeneity, and the nature of admissible true duration dependence, it is impossible to use longitudinal data to separate true duration dependence from spurious duration dependence. With regard to identifying true occurrence dependence, the situation is more favourable. From data on individual labour market histories generated in a stationary economic environment it is possible to use longitudinal data to separate true occurrence dependence from spurious occurrence dependence without having to resort to arbitrary assumptions. However, in a non-stationary external economic environment it is necessary to control for the non-stationarity, and to do this requires making some assumptions about the nature of the non-stationarity. However, in this case approximate tests for occurrence dependence can still be constructed.

The conditions required to identify true lagged duration dependence resemble those required in the standard econometric problem of estimating a distributed lag model in the presence of serially correlated errors. Knowledge of the correlations among lengths of successive spells of unemployment does not suffice to identify true lagged duration dependence. Such information must be supplemented either by *a priori* restrictions on the nature of admissible serial correlation and/or the length of the admissible lags, or else by using time varying exogenous variables as instrumental variables in a way to be described below (see Section II(b)).

Occurrence dependence is defined by comparisons of *marginal* exit time distributions for successive employment and unemployment spells controlling for heterogeneity. If the marginal exit time distributions are identical for successive spells of employment and for successive spells of unemployment, we say that there is no occurrence dependence in employment and unemployment respectively. Duration dependence is defined in terms of a particular characteristic of a marginal distribution: the hazard function. If the hazard function does not depend on the length of time in the spell, so that the distribution of exit time is exponential, we say that there is no duration dependence. Duration dependence for the waiting

time of a particular spell of an event is thus defined to exist if the marginal distribution for the waiting time is non-exponential. Absence of true lagged duration dependence is defined in terms of *independence of duration times* controlling for heterogeneity.

The presence of occurrence dependence is thus determined by comparing marginal distributions. Duration dependence is determined by a particular feature of the marginal distributions. Lagged duration dependence is determined by the nature of the statistical dependence or independence of the exit times of employment and unemployment spells.

Define $\mathbf{Z}(\tau)$ as a vector of exogenous variables at calendar time τ . $\theta(\tau)$ is a vector of unmeasured pure heterogeneity components at time τ . The distribution of $\theta(\tau)$ is assumed to be strictly stationary. Assume $\mathbf{Z}(\tau) = \bar{\mathbf{Z}}$ for all τ . These two assumptions define what we mean by a stationary environment. It is further assumed that the $\theta(\tau)$ components are independently distributed *across people*, and are drawn from a common distribution.

Write the joint distribution of employment and unemployment durations $t_{12}^{(1)}, \dots, t_{12}^{(n)}, t_{21}^{(1)}, \dots, t_{21}^{(n)}$ conditional on $\theta(\tau), \bar{\mathbf{Z}}$ for all τ as

$$(14) \quad F(t_{12}^{(1)}, \dots, t_{12}^{(n)}, t_{21}^{(1)}, \dots, t_{21}^{(n)} | \bar{\mathbf{Z}}, \theta(\tau), \text{all } \tau).$$

We use the shorthand expression “all τ ” for $\infty \geq \tau \geq 0$. This phrase is deleted when it is obvious from the context.

Equation (14) is a theoretical distribution of exit times from the first n spells of employment and unemployment. It is a marginal joint distribution of an infinite dimensional distribution of exit times from employment and unemployment with durations integrated out for spells with serial order greater than n . Economic hypotheses about occurrence, duration and lagged duration dependence are hypotheses about different features of this distribution which “holds constant” the intrusive effect of heterogeneity.

We assume that the data at our disposal comes from a panel sample of labour market histories in which all individuals have completed at least n spells of employment and unemployment so that we can ignore sampling issues. The problem of sample selection bias owing to incomplete intervals is discussed below. It is assumed that the exogenous variables $\bar{\mathbf{Z}}(\tau)$ can be observed, so that the available data can be summarized by the following distribution function.

$$(15) \quad G(t_{12}^{(1)}, \dots, t_{12}^{(n)}, t_{21}^{(1)}, \dots, t_{21}^{(n)} | \bar{\mathbf{Z}}, \text{for all } \tau) \\ = E_{\theta} [F(t_{12}^{(1)}, \dots, t_{12}^{(n)}, t_{21}^{(1)}, \dots, t_{21}^{(n)} | \bar{\mathbf{Z}}, \theta(\tau))].$$

This is the expectation of F with respect to the distribution of θ (i.e., unmeasured pure heterogeneity is “integrated out”).

Define $G^{(i)}$ as the *marginal distribution* of the i th random variable that appears in G . Thus $G^{(1)}$ is the marginal distribution of $t_{12}^{(1)}$. $G^{(n+1)}$ is the marginal distribution of $t_{21}^{(1)}$.

In a stationary external environment, the absence of occurrence dependence in employment is defined as

$$G^{(1)} = G^{(2)} = \dots = G^{(n)}.$$

The absence of occurrence dependence in unemployment is defined as

$$G^{(n+1)} = G^{(n+2)} = \dots = G^{(2n)}.$$

This definition of the absence of occurrence dependence can be generalized to allow only a subset of the marginal distributions to be the same.

The absence of *mean occurrence dependence* is the requirement that the mean durations of employment are identical across spells. The absence of mean occurrence dependence in unemployment requires equality of mean durations of unemployment across spells. One can clearly define second-moment or higher-moment occurrence dependence in an analogous fashion.

These are operational empirical definitions in the sense that one can, in principle, compare empirical marginal distributions of duration times or estimated means to perform a test. This definition of the absence of occurrence dependence is defined in terms of population distributions or moments, which can be estimated. A *structural* definition of no occurrence dependence is based on F , the joint distribution of duration times that “holds pure heterogeneity fixed”. Defining $F^{(i)}$ as the marginal distribution of the i th argument of F , the absence of *structural* occurrence dependence can be defined as

$$F^{(1)} = \dots = F^{(n)} \\ F^{(n+1)} = \dots = F^{(2n)}$$

provided that $\theta(\tau)$ is pre-specified to equal θ^* for all τ and that the equalities hold for each value of admissible θ^* . Given the stationarity assumption, the definition of the absence of structural occurrence dependence implies and is implied by the first definition. The same is true for the definitions of mean occurrence dependence. One can also define the absence of occurrence dependence for subsets of the spells. The key notion in the concept of no occurrence dependence is that the marginal distributions of exit times from employment and unemployment states are stationary across successive spells. Note that under our assumptions for successive equispaced intervals of calendar time, the definition of no occurrence dependence implies equal mean numbers of counts of unemployment and employment within each interval.

Duration dependence is defined by the existence of a non-constant hazard function within a spell; i.e., we say that there is no duration dependence in employment spell l if

$$(a) \quad \frac{dG^{(l)}(t_{12}^{(l)})}{dt_{12}^{(l)}} \frac{1}{1 - G^{(l)}(t_{12}^{(l)})} = m^{(l)}$$

where $m^{(l)}$ does not depend on $t_{12}^{(l)}$. As previously noted, this definition implies and is implied by an exponential duration time distribution. This definition is empirically verifiable using estimated hazard functions.

A structural definition of duration dependence in employment spell l is that the structural hazard function is independent of duration time $t_{12}^{(l)}$, i.e., that

$$(b) \quad \frac{dF^{(l)}(t_{12}^{(l)})}{dt_{12}^{(l)}} \frac{1}{1 - F^{(l)}(t_{12}^{(l)})} = n^{(l)}$$

where $n^{(l)}$ does not depend on $t_{12}^{(l)}$. Unfortunately, even in a stationary environment evidence that an empirical hazard function is independent of $t_{12}^{(l)}$ is not evidence that the structural hazard function is independent of $t_{12}^{(l)}$. It is possible to observe empirical hazard functions that satisfy condition (a) even though the structural

hazard function does not satisfy condition (b).⁹ Conversely, as demonstrated by Silcock (1954), it is possible to observe empirical hazard functions that do not satisfy condition (a) even though the structural hazard function satisfies condition (b). Without restricting the nature of the admissible functional forms for $F^{(l)}$, and without restricting the nature of the admissible pure heterogeneity distributions, it is not possible to test for structural duration dependence. This result implies that any empirical test for or measurement of structural duration dependence can be conducted only with strong *a priori* assumptions. In most cases, such assumptions usually cannot be justified by an appeal to economic theory.

The basic problem is that from the data we can estimate $G^{(l)}(t_{12}^{(l)})$, but $G^{(l)}(t_{12}^{(l)}) = E_{\theta} F^{(l)}(t_{12}^{(l)})$. From the estimated empirical distribution function we cannot separate out $F^{(l)}$ from an arbitrary distribution of pure heterogeneity without imposing severe restrictions on the admissible distributions.

True lagged duration dependence is said not to exist for exit times from the employment state if, conditional on $\theta(\tau)$ and $\mathbf{Z}$, for all τ , $t_{12}^{(1)}, \dots, t_{12}^{(n)}$, are mutually independent random variables. Thus there is no true lagged duration dependence in employment duration if

$$\begin{aligned} F(t_{12}^{(1)}, \dots, t_{12}^{(n)}, \infty, \dots, \infty | \bar{\mathbf{Z}}(\tau), \theta(\tau), \text{all } \tau) \\ = F^{(1)}(t_{12}^{(1)} | \theta(\tau), \mathbf{Z}(\tau)) \dots F^{(n)}(t_{12}^{(n)} | \theta(\tau), \bar{\mathbf{Z}}). \end{aligned}$$

Similarly, there is no lagged duration dependence in unemployment durations if

$$\begin{aligned} F(\infty, \dots, \infty, t_{21}^{(1)}, \dots, t_{21}^{(n)} | \bar{\mathbf{Z}}, \theta(\tau), \text{all } \tau) \\ = F^{(n+1)}(t_{21}^{(1)} | \theta(\tau), \bar{\mathbf{Z}}) \dots F^{(2n)}(t_{21}^{(n)} | \theta(\tau), \bar{\mathbf{Z}}). \end{aligned}$$

There is no true lagged duration dependence at all if

$$\begin{aligned} F(t_{12}^{(1)}, \dots, t_{12}^{(n)}, t_{21}^{(1)}, \dots, t_{21}^{(n)} | \bar{\mathbf{Z}}, \theta(\tau), \text{all } \tau) \\ = \prod_{j=1}^n F^{(j)}(t_{12}^{(j)} | \theta(\tau), \bar{\mathbf{Z}}) \prod_{j=1}^n F^{(j+n)}(t_{21}^{(j)} | \theta(\tau), \bar{\mathbf{Z}}). \end{aligned}$$

Lagged duration dependence is defined to exist if the conditional distribution of $t_{ij}^{(l)}$ given lagged $t_{ij}^{(l)}$ and $\theta(\tau)$ and $\bar{\mathbf{Z}}$ is not the same as the conditional distribution of $t_{ij}^{(l)}$ given $\theta(\tau)$ and $\bar{\mathbf{Z}}$.

Unless we somehow control for the pure heterogeneity components $\theta(\tau)$, so that we directly estimate distribution function (14), the distribution of sample exit times is generated by distribution (15). Independence of the random variables generated from G does not imply independence of the random variables generated by F . The converse is also true. Accordingly, tests for independence based on sample data which do not correct for unobserved heterogeneity are uninformative about true lagged duration dependence. In Section II we present some regression tests for special forms of true lagged duration dependence that permit us to control for unobserved components of heterogeneity.

Occurrence dependence does not imply and is not necessarily implied by lagged duration dependence. However, in special cases lagged duration dependence may imply occurrence dependence.

For example, suppose that true lagged duration dependence is of the special autoregressive form

$$\ln t_{ij}^{(l)} = \beta \ln t_{ij}^{(l-1)} + \varepsilon_{ij}^{(l)}, \quad 0 < \beta < 1$$

where $\varepsilon_{ij}^{(l)}$ is an independently identically distributed mean zero random variable assumed to be distributed independently of $\ln t_{ij}^{(l-1)}$. For arbitrary initial conditions, the marginal distribution of $\ln t_{ij}^{(l)}$ is not the same as the marginal distribution of $\ln t_{ij}^{(l-1)}$.¹⁰ Because $\beta \neq 0$, there is lagged duration dependence and occurrence dependence. However, if the process starts from an equilibrium distribution for $\ln t_{ij}^{(0)}$, successive marginal distributions of $\ln t_{ij}^{(l)}$ ($l = 1, \dots, L$) are identical so there is no occurrence dependence but there is lagged duration dependence.

In a non-stationary environment, the $\mathbf{Z}(\tau)$ and the distribution of $\theta(\tau)$ will be functions of time. Without some prior restrictions on the nature¹¹ of how the $\mathbf{Z}(\tau)$ and $\theta(\tau)$ determine F , and how the distribution of $\theta(\tau)$ depends on time, no test for structural occurrence dependence is possible, because the empirical duration distributions will shift across spells even in the absence of structural occurrence dependence. However, an *approximate* test for mean occurrence dependence can be constructed by regressing mean durations on the exogenous variables (including τ). Via regression, it is possible to control *approximately* for the effect of non-stationarity in the environment on conditional duration times, and so generate an *approximate* test for mean occurrence dependence. By “approximate” we mean that we can use polynomial approximations for the conditional regression functions with a high degree of accuracy. This point is developed further in Section II(b). There it is noted that non-stationarity in the $\mathbf{Z}(\tau)$ *aids* in the identification of lagged duration dependence.

II. METHODS OF TESTING FOR STATE DEPENDENCE

This section presents two different procedures for testing for true state dependence. The first procedure is based on conditional distributions of exit times from a state. The second procedure is based on regression analysis of duration intervals. A third procedure, based on maximum likelihood methods, is developed in companion papers (Heckman and Borjas, 1979; Flinn and Heckman, 1980). As the procedures are presented, their relative merits are assessed.

(a) Tests based on conditional exit times

The basic idea in this sub-section is due to Bates and Neyman (1952), but it also appears in various places in the literature (see Cox and Lewis, 1966, p. 21).

Assume no true duration or lagged duration dependence, and further assume a stationary environment. Under the null hypothesis of no occurrence dependence, the rate of transition out of the j th employment spell for individual i is $a_{12}^{(j)}(i) = a_{12}(i)$ for all j . Measuring the individual's labour market history by an employment clock, so that in terms of this clock time spent in unemployment is treated as a point event which takes no time at all, under the null hypothesis of no occurrence dependence in employment, the density of duration times of completed employment spells of length $t_1, t_2, \dots, t_{L-1}, t_L$ in an interval of time in employment, $(0, K]$, is

$$a_{12}(i) \exp\{-a_{12}(i)t_1\} a_{12}(i) \exp\{-a_{12}(i)t_2\} \dots \\ \exp\{-a_{12}(i)(K - t_L - t_{L-1} - \dots - t_1)\} = (a_{12}(i))^L \exp\{-a_{12}(i)K\}.$$

where the final term on the left-hand side is the probability that individual i is in an uncompleted $(L + 1)$ st spell at the end of the interval of K units of employment

time. In order to write the density in this fashion it is necessary to assume that explanatory variables and heterogeneity for individual i do not change across spells.

The probability of L completed employment spells in the interval $(0, K]$ is given from the Poisson distribution as

Prob (L events of leaving employment in interval $(0, K]$)

$$= \exp\{-a_{12}(i)K\} \frac{(a_{12}(i)K)^L}{L!}.$$

The conditional distribution of exit times for an individual who experiences L completed spells of employment in interval $(0, K]$ is

$$(16) \quad f(t_1, \dots, t_L | L \text{ completed spells in } (0, K]) \\ = \frac{(a_{12}(i))^L \exp\{-a_{12}(i)K\}}{\exp\{-a_{12}(i)K\} \frac{(a_{12}(i)K)^L}{L!}} = \frac{L!}{K^L}.$$

For $L = 1$, the conditional density of exit times is a univariate uniform density *that does not depend on individual pure heterogeneity parameters*. By choosing a suitable type of conditioning, it is possible to eliminate the intrusive effect of heterogeneity.

From the analysis of Bates and Neyman it is possible to demonstrate that, under the alternative hypothesis of occurrence dependence ($a_{12}^{(m)}(i) \neq a_{12}^{(m+1)}(i)$), the conditional density of the first completed employment spell in the interval $(0, K]$, t_1 , given m spells of employment up through time 0, and only one completed employment spell in $(0, K]$, is given by

$$(17) \quad \frac{\psi_{12}^{(m)}(i) \exp\{\psi_{12}^{(m)}(i)t_1\}}{\exp\{\psi_{12}^{(m)}(i)K\} - 1}$$

where $\psi_{12}^{(m)}(i) = a_{12}^{(m+1)}(i) - a_{12}^{(m)}(i)$. Under the null hypothesis of no occurrence dependence ($\psi_{12}^{(m)}(i) = 0$), the distribution of t_1 is uniform. A more general class of alternatives is presented in our companion paper.

Using conditional exit time distributions, it is possible to construct a simple test of the hypothesis of no occurrence dependence—a test for a uniform distribution. The Kolmogorov–Smirnov or χ^2 statistics can be used to test the hypothesis.

This test is not robust with respect to changes in components of heterogeneity across spells. To demonstrate this point, write

$$(18) \quad a_{12}^{(m)}(i) = \beta_{12}^{(m)} \mathbf{Z}^{(m)}(i) + f^{(m)}(i)$$

where $\mathbf{Z}^{(m)}(i)$ is a vector of measured exogenous explanatory variables for individual i in spell m , and where $f^{(m)}(i)$ is a portmanteau variable of unmeasured omitted variables assumed to be distributed independently of $\mathbf{Z}^{(m)}(i)$. With these definitions, $\psi_{12}^{(m)}(i)$ may be written as

$$(19) \quad \psi_{12}^{(m)}(i) = \beta_{12}^{(m+1)} \mathbf{Z}^{(m+1)}(i) - \beta_{12}^{(m)} \mathbf{Z}^{(m)}(i) + f^{(m+1)}(i) - f^{(m)}(i).$$

Evidence against a uniform distribution of conditional exit times is consistent with any or all of the following explanations:

- (a) $\beta_{12}^{(m)} \neq \beta_{12}^{(m+1)}$
- (b) $\mathbf{Z}^{(m)}(i) \neq \mathbf{Z}^{(m+1)}(i)$
- (c) $f^{(m)}(i) \neq f^{(m+1)}(i)$.

The first explanation is due to occurrence dependence. In the presence of such state dependence, the same characteristics have different effects on the probability of experiencing unemployment in different employment spells. The second explanation is due to changes in *exogenous* explanatory variables between the m th and $(m+1)$ st spells, which by definition are *not* related to the employment duration process. Hence rejection of a uniform conditional exit time distribution because of (b) does not imply the presence of occurrence dependence.¹² The third explanation is due to changes in unmeasured variables across spells. Whether evidence against a uniform distribution due to this source is evidence for or against occurrence dependence is problematic. For example, if $f^{(m+1)}(i) = \eta^{(m+1)} \phi(i) + u^{(m+1)}(i)$ where $\phi(i)$ is a permanent component constant over all spells, and $u^{(m+1)}(i)$ is distributed independently of all other values of $u^{(j)}(i)$, ($j \neq m+1$), $f^{(m+1)}(i) - f^{(m)}(i) = (\eta^{(m+1)} - \eta^{(m)}) \phi(i) + u^{(m+1)}(i) - u^{(m)}(i)$. If $\eta^{(m+1)} = \eta^{(m)}$, empirical evidence against uniformity due to explanation (c) would not be evidence in favour of the hypothesis of occurrence dependence, assuming that the transitory components are not generated by previous outcomes of the process. On the other hand, if $\eta^{(m+1)} \neq \eta^{(m)}$, evidence against a uniform distribution from explanation (c) would be evidence in support of the hypothesis of occurrence dependence.¹³ In this case, explanation (c) is on the same logical footing as explanation (a) except that $\phi(i)$ is an unobserved variable while the elements of $\mathbf{Z}^{(m)}(i)$ are observed.

The upshot of this discussion is that failure to accept the hypothesis of a uniform conditional distribution of duration times is not informative. Unless heterogeneous components remain constant over spells for the individual, rejection of the uniformity hypothesis does not necessarily imply the presence of occurrence dependence. However, acceptance of the hypothesis implies the absence of occurrence dependence.

It is important to recall that throughout this sub-section we have assumed that there is no duration dependence or lagged duration dependence. If there is duration dependence (within or across spells), but no occurrence dependence, distributions (16) and (17) will not be uniform.¹⁴ Accordingly, a test for a uniform conditional waiting time distribution is also a test of the null hypothesis of no duration dependence within spells or lagged duration dependence across spells.

(b) Regression tests

This sub-section develops regression approaches to testing for occurrence, duration and lagged duration dependence. Regression procedures are more informative than the Bates–Neyman procedure. With them it is possible to eliminate or control the contaminating effect of variables that change across spells of employment and unemployment. The regression approach has the additional advantage that it yields point estimates of duration, occurrence and lagged duration dependence parameters. Moreover, very general types of unmeasured

heterogeneity may be introduced into the model at low computational cost. In this regard, the regression approach is more attractive than maximum likelihood procedures.

The basic idea underlying the regression test for occurrence dependence is that, if there is no such dependence, expected values of durations of successive spells of employment (or unemployment) should be equal provided that the effects of variables that change across spells are controlled. Regressing the duration of completed spells of employment (or unemployment) on spell-specific explanatory variables controls for the effects of changes in exogenous variables on the durations of successive spells. This test for occurrence dependence is quite general and is consistent with alternative specifications of duration dependence. Estimates of duration dependence are much more sensitive to specific choices of functional forms for duration equations, and hence are less convincing because there is usually no good reason for preferring one functional form over another. Tests for occurrence dependence require fewer *a priori* assumptions than tests for duration dependence. Tests for lagged duration dependence are formally equivalent to tests for the existence of distributed lag responses in a model with serially correlated errors, so that it is possible to use well-developed arguments about identification in distributed lag models to identify lagged duration dependence effects.

Our analysis of the regression approach is conducted for the general Weibull and log-normal models with parameterizations given in equations (11a) and (11b). We adopt the specification of heterogeneity proposed in equations (12a) and (12b). For the l th spell of event i the Weibull regression model is in the form of a distributed lag relationship

$$(20a) \quad \ln t_{ij}^{(l)} = \frac{1}{\gamma_{ij}^{(l)}} \left\{ B_1 + \beta_{ij}^{(l)} \mathbf{Z}_{ij}^{(l)} + \sum_{k=1}^{l-1} \eta_{1l}^{(l)}(k) \ln t_{ij}^{(l-k)} + \sum_{k=0}^{l-1} \eta_{2l}^{(l)}(k) \ln t_{ij}^{(l-k)} \right\} \\ + \frac{1}{\gamma_{ij}^{(l)}} V_{ij}^{(l)} + W_{ij}^{(l)} \quad i, j = 1, 2, i \neq j$$

where

$$E(V_{ij}^{(l)}) = 0 \quad E(V_{ij}^{(l)})^2 = \sigma_l^{(l)}, \quad E(V_{ij}^{(l)} V_{i'j'}^{(l')}) = \sigma_{l,i'}^{(l,l')}, \quad l \neq l' \\ E(W_{ij}^{(l)}) = 0 \quad E(W_{ij}^{(l)})^2 = B_2 / (\gamma_{ij}^{(l)})^2, \quad E(W_{ij}^{(l)} W_{i'j'}^{(l')}) = 0 \\ E(W_{ij}^{(l)} V_{i'j'}^{(l')}) = 0 \quad \text{for all } l, l', l \neq l', i, i', j, j' = 1, 2, i \neq j, i' \neq j'.$$

where B_1 and B_2 are known constants.¹⁵

For the l th spell of event i , the log-normal regression model is

$$(20b) \quad \ln t_{ij}^{(l)} = \tilde{\beta}_{ij}^{(l)} \mathbf{Z}_{ij}^{(l)} + \sum_{k=1}^{l-1} \tilde{\eta}_{1l}^{(l)}(k) \ln t_{ij}^{(l-k)} + \sum_{k=0}^{l-1} \tilde{\eta}_{2l}^{(l)}(k) \ln t_{ij}^{(l-k)} \\ + \tilde{V}_{ij}^{(l)} + N_{ij}^{(l)}, \quad i, j = 1, 2, i \neq j$$

where

$$E(\tilde{V}_{ij}^{(l)}) = 0 \quad E(\tilde{V}_{ij}^{(l)})^2 = \tilde{\sigma}_l^{(l)}, \quad E(\tilde{V}_{ij}^{(l)} \tilde{V}_{i'j'}^{(l')}) = \tilde{\sigma}_{l,i'}^{(l,l')} \\ E(N_{ij}^{(l)}) = 0 \quad E(N_{ij}^{(l)})^2 = \tilde{\sigma}_l^{(l)}, \quad E(N_{ij}^{(l)} N_{i'j'}^{(l')}) = 0$$

for all $l, l', l \neq l', i, i', j, j' = 1, 2, i \neq j, i' \neq j'$. As before, tildes are used to distinguish log-normal from Weibull parameters. Note that the error structure for hetero-

geneity components $V_{ij}^{(l)}$ and $\tilde{V}_{ij}^{(l)}$ is left unspecified and so is consistent with a variety of assumptions about serial correlation in unmeasured components.

For both the Weibull and log-normal models, lagged duration dependence is defined to exist if at least some of the coefficients of the lagged duration variables are non-zero. Mean occurrence dependence is defined to exist if the parameters of the regression function change for successive spells of event i . As previously noted, the log-normal model imposes a special form of duration dependence. In the Weibull model, duration dependence is defined to exist in spell l of event i if $\gamma_{ij}^{(l)} \neq 1$.

To simplify the exposition we discuss tests for occurrence, duration and lagged duration dependence in order. For the moment, assume no lagged duration dependence. Under the null hypothesis of no occurrence dependence the mean duration of successive spells of event i is the same, so long as the control variables, $\mathbf{Z}_{ij}^{(l)}$, stay constant across spells. Shifts in mean duration across spells owing to exogenous factors do not give rise to occurrence dependence. A definition of occurrence dependence based solely on the means of the distributions of exit times is termed "mean occurrence dependence". A more general definition of the absence of occurrence dependence requires that the duration time distributions remain the same across successive spells except for shifts in exogenous variables. In the log-normal framework, the more general definition imposes the additional requirement that $\tilde{\sigma}_{ij}^{(l)} = \tilde{\sigma}_{ij}^{(l')}$, for all l and l' , and that the distribution of heterogeneity ($\tilde{V}_{ij}^{(l)}$) is stationary across spells. In the Weibull framework, the more general definition imposes the additional requirement that the distribution of $V_{ij}^{(l)}$ is stationary across successive spells.

A simple way to test for mean occurrence dependence is to estimate the regression functions (20a) and (20b) for each successive spell of event i (recall that for the moment we are ignoring lagged duration dependence). A simple test of equality of slope and intercept coefficients of the conditional regression function for mean duration in successive spells is a test of the null hypothesis of no occurrence dependence. Note that, even if there are *no* regressor variables in the model, or if there are regressor variables that are invariant across spells, a test of the null hypothesis of no mean occurrence dependence can be performed using simple differences in mean durations across successive spells.¹⁶ In fact, for reasons discussed in the next sub-section, it is sometimes advantageous to have access to samples for which no exogenous variables are relevant so that certain technical difficulties that arise from introducing explanatory variables can be avoided. The presence or absence of exogenous variables in the model is irrelevant for defining or measuring occurrence dependence, and their absence simplifies the testing procedure.

An important feature of this test for occurrence dependence is that exactly the same test procedure is used whether the Weibull or log-normal model is used. As previously noted, these two models generate very different types of duration dependence. Thus our test for mean occurrence dependence is independent of the form of the duration dependence.

This point is more general. For *any* form of within-spell duration dependence, a test of the hypothesis of no mean occurrence dependence can be based on a test of equality of conditional mean duration regression functions for successive spells. The conditional mean duration functions can be approximated to a high degree of accuracy. Without regressors, no approximation is involved. Thus our test for

mean occurrence dependence does not require strong *a priori* assumptions about the nature of duration dependence.

A test for general occurrence dependence is essentially a test of the stationarity of duration time distributions across successive spells (controlling for changes in the exogenous variables). In the absence of exogenous variables, it is possible to fit empirical duration time distributions for each spell (using pooled duration data from individuals) and compare empirical distributions across spells. In the absence of occurrence dependence, the empirical distributions should be the same across successive spells. If exogenous variables are present, and if one is prepared to adopt a log-normal assumption for duration times, a test for general occurrence dependence is a test for mean occurrence dependence and a test for equality of variances across spells. If one is not prepared to adopt a log-normal duration time assumption, a more general parametric or non-parametric density may be fit, and fitted distributions across spells can be compared (controlling for changes in the exogenous variables). As in the case of testing for mean occurrence dependence, it is not necessary to impose strong distributional assumptions in order to test for general occurrence dependence.

At the cost of imposing considerable further structure on the nature of heterogeneity, and the nature of duration dependence, it is possible to test for and estimate duration dependence. To see this, consider the Weibull model. If $\gamma_{ij}^{(l)} = 1$, there is no duration dependence in the l th spell of event i . Otherwise, there is duration dependence. Unless some structure is imposed on the covariance structure of heterogeneity component $V_{ij}^{(l)}$, it is not possible to identify duration dependence parameter $\gamma_{ij}^{(l)}$.¹⁷

Suppose that we specialize the nature of heterogeneity so that $V_{ij}^{(l)} = C_{ij}\phi$, where ϕ is a permanent component with $E(\phi) = 0$, $E(\phi^2) = \sigma_\phi^2$. The error structure of the log duration regression equations then has a special one-factor structure consisting of the sum of the independently distributed component $W_{ij}^{(l)}$ and the permanent component, i.e.

$$\frac{C_{ij}\phi}{\gamma_{ij}^{(l)}} + W_{ij}^{(l)}.$$

The full regression model can then be written as

$$(21) \quad \ln(t_{ij}^{(l)}) = \frac{1}{\gamma_{ij}^{(l)}} (B_1 + \beta_{ij}^{(l)} \mathbf{Z}_{ij}^{(l)} + C_{ij}\phi) + W_{ij}^{(l)}, \quad l = 1, \dots, L.$$

Provided that we have data on two or more completed spells of event i , (say, l and l') from a longitudinal sample, all of the parameters of equation (21) can be estimated. Regression yields estimates of $\beta_{ij}^{(l)}/\gamma_{ij}^{(l)}$, except for the constant term which includes $B_1/\gamma_{ij}^{(l)}$. From the residual variance from each regression, one can estimate

$$(22a) \quad \frac{C_{ij}^2 \sigma_\phi^2}{(\gamma_{ij}^{(l)})^2} + \frac{B_2}{(\gamma_{ij}^{(l)})^2}, \quad l = 1, \dots, L.$$

The covariance between the disturbances in spells l and l' yields an estimate of

$$(22b) \quad \frac{C_{ij}^2 \sigma_\phi^2}{\gamma_{ij}^{(l)} \gamma_{ij}^{(l')}}.$$

Assuming that the number of individuals in the panel is large, sample moments consistently estimate population moments, and equations (22a) and (22b) can be solved for unique estimates of $C_{ij}^2 \sigma_\phi^2$, $\gamma_{ij}^{(j)}$ and $\gamma_{ij}^{(l')}$.¹⁸ Given the estimates of $\gamma_{ij}^{(j)}$ and $\gamma_{ij}^{(l')}$, one can identify all the components of $\beta_{ij}^{(j)}$ and $\beta_{ij}^{(l')}$.

These estimates of duration dependence are sensitive to measurement error in the dependent variable (or components of unobserved heterogeneity) that is independently distributed across spells. (Permanent components of measurement error are absorbed in ϕ leading to an upward bias in the estimate of $C_{ij}^2 \sigma_\phi^2$ but no bias in the other coefficients.) It is straightforward to show that such measurement error generates downward-biased estimates of the duration dependence parameters obtained from the procedure just described.

One way to achieve estimates of the *ratio* of duration dependence parameters in the presence of independently distributed measurement error is to note that equation (21) may be written in autoregressive form. Since

$$C_{ij}\phi = \gamma_{ij}^{(l)} \ln(t_{ij}^{(l)}) - B_1 - \beta_{ij}^{(l)} \mathbf{Z}_{ij}^{(l)} - W_{ij}^{(l)}$$

one can substitute for ϕ in the equation for $\ln t_{ij}^{(l')}$ and write

$$\ln t_{ij}^{(l')} = \frac{1}{\gamma_{ij}^{(l')}} (-\beta_{ij}^{(l)} \mathbf{Z}_{ij}^{(l)} + \beta_{ij}^{(l')} \mathbf{Z}_{ij}^{(l')}) + \frac{\gamma_{ij}^{(l)}}{\gamma_{ij}^{(l')}} \ln t_{ij}^{(l)} - \frac{W_{ij}^{(l)}}{\gamma_{ij}^{(l')}} + W_{ij}^{(l')}.$$

Clearly, $\ln t_{ij}^{(l')}$ is correlated with $W_{ij}^{(l)}$ so that ordinary regression estimates of the parameters of this equation are not consistent. Provided that one has data on a third spell of employment, l'' , one can use $\ln t_{ij}^{(l'')}$ as a valid instrument for $\ln t_{ij}^{(l')}$ and so achieve consistent estimates of $\gamma_{ij}^{(l)}/\gamma_{ij}^{(l'')}$.¹⁹ Using the standard asymptotic theory (assuming the number of individual event histories becomes large), one can test whether duration dependence changes across spells, but one cannot, directly, estimate duration dependence parameters.

It is important to note that estimates of $\gamma_{ij}^{(l)}$ or $\gamma_{ij}^{(l)}/\gamma_{ij}^{(l')}$ are quite sensitive to the specification of the covariance of unmeasured heterogeneity across spells. In order to secure estimates of these parameters, one must impose strong (and usually arbitrary) assumptions on the nature of the distribution of these components. With heterogeneity freely correlated across spells, no meaningful estimate of duration dependence is possible.

The prospects for obtaining meaningful duration dependence parameters worsen when we entertain the possibility that log employment duration is normally distributed and not Weibull distributed. As previously noted, the test for mean occurrence dependence proposed above remains valid in this case. But estimates of duration dependence based on the Weibull assumption are meaningless.²⁰ The log normal distribution *imposes* a special sort of duration dependence. A test between the log-normal and Weibull distribution based on an analysis of regression residuals is sensitive to assumptions made about the distribution of unmeasured components of heterogeneity. Usually there is no basis for preferring one distribution over another.

The upshot of this discussion is that regression tests for mean occurrence dependence are robust with respect to quite general specifications for heterogeneity and do not require strong *a priori* assumptions. Tests for duration dependence are quite fragile and require arbitrary assumptions that are difficult to justify. Tests for general occurrence dependence in the presence of general forms of heterogeneity remain to be rigorously developed, although we have proposed

some intuitive procedures (comparing empirical exit time distributions for successive spells).

A regression test for lagged duration dependence can also be constructed. In equation (20a), one requires consistent estimators for $\eta_{1i}^{(l)}(k)$ and $\eta_{2i}^{(l)}(k)$, $k = 1, \dots, l - 1$. If the heterogeneity components $V_{ij}^{(l)}$ are uncorrelated across episodes, ordinary least squares can be used to estimate the lag coefficients up to a factor of proportionality ($\gamma_{ij}^{(l)}$). However, if heterogeneity components are correlated, these estimated coefficients are biased, even in large samples.

The problem of testing for lagged duration dependence variables in this case is analogous to the problem of testing for lagged dependent variables in the presence of serial correlation (see Griliches, 1967, especially p. 35). If there are exogenous variables that change across spells, one can use the lagged exogenous variables as instruments for the lagged duration variables.²¹ One can also use restrictions on the order of the serial correlation in the heterogeneity components to identify the model. However, as noted by Hatanaka (1975) and others, if the exogenous variables in the model stay constant across spells, and the order of the serial correlation in the heterogeneity components is unknown, it is impossible to estimate consistently the $\eta_{1i}(p)$ and $\eta_{2i}(q)$, and therefore it is impossible to distinguish lagged duration dependence from heterogeneity.

We conclude this sub-section with a simple procedure for testing for mean occurrence dependence. Log duration in spell l is written as

$$\ln t_{ij}^{(l)} = \mathbf{b}_{ij}^{(l)} \mathbf{Z}_{ij}^{(l)} + R_{ij}^{(l)}, \quad l = 1, \dots, L$$

where $R_{ij}^{(l)}$ is a mean zero disturbance term, with finite second moments. The disturbance can be generated by a general distribution of heterogeneity, and interspell covariance is arbitrary. This regression equation can be justified by invoking log-normal or Weibull duration distributions, or by assuming it is an approximation to the conditional expectation of log duration in a more general model. ($\mathbf{Z}_{ij}^{(l)}$ may contain polynomials in the explanatory variables.)

Define $\Delta_{ij}^{(l,l')} = \mathbf{Z}_{ij}^{(l)} - \mathbf{Z}_{ij}^{(l')}$. Then

$$(23) \quad \ln t_{ij}^{(l)} - \ln t_{ij}^{(l')} = \mathbf{b}_{ij}^{(l')} \Delta_{ij}^{(l,l')} + (\mathbf{b}_{ij}^{(l')} - \mathbf{b}_{ij}^{(l)}) \mathbf{Z}_{ij}^{(l)} + R_{ij}^{(l)} - R_{ij}^{(l')}.$$

Least squares can be used to estimate the parameters in equation (23).²² Invoking standard asymptotic arguments to justify sampling distributions, one can use the estimated coefficients associated with regressor vector $\mathbf{Z}_{ij}^{(l)}$ to test for the presence of occurrence dependence by testing the null hypothesis that $\mathbf{b}_{ij}^{(l)} \neq \mathbf{b}_{ij}^{(l')}$. Rejection of this null hypothesis is rejection of the hypothesis of no mean occurrence dependence.

(c) *Limitations of the regression procedure*

There are two principal limitations of the regression procedure. The first limitation is that it is ill-equipped to deal with censored observations, and the bias that results from censoring may be severe. The second limitation is that the regression framework does not lend itself to the ready introduction of explanatory variables, such as age, that change within a sample. Both of these topics are discussed at length in Heckman and Borjas (1979) and Flinn and Heckman (1980). Here, we simply indicate the problems.

In short panels, censoring bias may be an important phenomenon. To illustrate the problem in a simple way, consider estimation of the coefficients of the

equation for the first employment spell. To avoid notational clutter, we suppress the subscript denoting a first spell. The log duration equation is

$$(24) \quad \ln t_{12} = \mathbf{b}_{12} \mathbf{Z}_{12} + R_{12}.$$

Suppose that the panel data at our disposal is limited to T periods, and T is sufficiently short that a significant fraction of the observations are still in their first spell of employment by time T . Fitting regressions only on the observations with completed spells will lead to biased estimators since we are imposing the condition that $t_{12} \leq T$. The distribution of the composite error term, R_{12} , is now a conditional distribution which depends on $\mathbf{Z}_{12}$ and $\mathbf{b}_{12}$. In general, the mean of the disturbances will be non-zero, and will depend on $\mathbf{Z}_{12}$ and the parameters of the model. Hence regression estimators of equation (24) fit on the censored sample will be biased unless some allowance is made for the conditioning required to generate the sample used for regression analysis.

For example, let $\mathbf{Z}_{12}$ be a scalar regressor, and let b_{12} be positive. Individuals with large values of $\mathbf{Z}_{12}$ will be under-represented in the sample of individuals with completed spells. For people with high values of $\mathbf{Z}_{12}$ to be in the sample they must have, on average, low values of the disturbance R_{12} so that regression estimators of b_{12} derived from censored samples are downward-biased.

For fixed T , and fixed values of b_{12} and $\mathbf{Z}_{12}$ across spells, the bias that arises from using censored samples becomes larger the higher the serial order of the spell analysed by regression analysis. The reason for this is simple: the sample becomes increasingly unrepresentative as the serial order of the spell analysed increases. The impact of censoring on values of b_{12} estimated from each spell will be to manufacture apparent occurrence dependence solely as a consequence of the sampling scheme.

It is important to note that the Bates–Neyman procedure is not affected by censoring because the conditioning used to perform the test eliminates the effect of heterogeneity in the sample. This is a major advantage of that procedure in comparison with the regression procedure.

Another limitation of the regression procedure is that it does not readily lend itself to the introduction of time-varying explanatory variables. The functional form of the expected value of log duration in a spell changes in the presence of such variables. *Ad hoc* expedients, such as using average values of time-varying explanatory variables within a spell, can generate bias. For example, suppose one used average age *within* a spell as an explanatory variable in a sample of individuals initially of the same age. The longer the spell, the greater the average age. Hence one could manufacture a spurious positive duration–age relationship by this *ad hoc* procedure. As another example, use of age at the *onset* of subsequent spells, raises similar simultaneity issues. Individuals of identical age at the onset of a process will be of different ages at the onset of subsequent spells. Age at the onset of later spells is determined, in part, by heterogeneity components in the earlier spells. If these components are correlated over spells, the age at onset in subsequent spells is correlated with the unobserved variable.

A final disadvantage of the regression procedure in the presence of time-varying exogenous variables is that, in order to compute the conditional mean duration for a spell, one must use *all* values of the exogenous variables *after* the spell commences in computing the mean, including values of the exogenous

variables after the spell has terminated. This topic is discussed extensively in our companion papers.

(d) *Maximum likelihood methods*

The essential ingredients required to estimate our model by the method of maximum likelihood have already been given in Section II. Using equations (5), (11) and (12), and specifying a distribution for heterogeneity, one can write out the probability of a particular sample outcome. Assuming random sampling, the log of the product of the density for all sample patterns is the sample log likelihood. A full discussion of the likelihood function and the large sample properties of estimators derived from it is given in our companion papers. We consider several density functions for heterogeneity, including the gamma and the normal densities.

Explanatory variables that change within each spell can be introduced into the model using densities of the sort presented in equation (13). Details of this procedure are found in our companion papers. For the Weibull case, one can use the information matrix derived from the sample likelihood to test the hypothesis of no mean occurrence dependence ($\beta_{lj}^{(l)} = \beta_{lj}^{(l')}$, $\gamma_{lj}^{(l)} = \gamma_{lj}^{(l')}$, $\beta_{li}^{(l)} = \beta_{li}^{(l')}$, $\gamma_{li}^{(l)} = \gamma_{li}^{(l')}$, $\eta_{li}^{(l)}(p) = \eta_{li}^{(l')}(p)$, $\eta_{2li}^{(l)}(p) = \eta_{2li}^{(l')}(p)$ for all $l, l', l \neq l'$). Similar tests can be constructed for the log-normal case.

Estimation by maximum likelihood enables us to introduce time-varying explanatory variables, and to avoid censoring problems. Its principal cost is that we must now specify the distribution of the unobserved components of heterogeneity—something not required in either the regression procedure or the Bates–Neyman procedure. In addition, unlike the regression test for mean occurrence dependence, tests for occurrence dependence based on maximum likelihood require one to specify the specific form of the duration dependence.

III. EMPIRICAL RESULTS

This section presents empirical results obtained from application of some of the tests proposed in this paper to data on monthly labour market histories available from the National Longitudinal Surveys of Young Men (NLS).²³ During the years 1969–1971 the NLS compiled a detailed work history for a random sample of young men. This history includes a complete listing of jobs held in those years as well as a systematic accounting of time not spent at work. From these data it is possible to construct a monthly index indicating the individual's labour force status over the period.

We restrict our analysis to a sub-sample of white men who have a high school diploma, who were interviewed in the 1969, 1970 and 1971 surveys, who completed their education in 1969, and who did not continue their education in the autumn of 1969. Thus we analyse the work history of young males for a 30-month period after completion of high school. By focusing on young men just emerging from school, we avoid, in large part, the problem of initial conditions and the problem of missing relevant pre-sample labour market histories that have received considerable attention in the recent literature on estimating discrete-time stochastic processes (Heckman, 1978). For most individuals in our sample, the first job after high school graduation is their first full-time job. The length of our panel sample is determined by the fact that most men graduated from high school

in May–June 1969, and the continuous monthly work history can be accurately compiled up to November–December 1971. Our sample is composed of the 122 individuals who satisfied the aforementioned sample selection rules.

(a) *Results from the Bates–Neyman procedure*

We seek to determine whether unemployment affects the future probability of becoming unemployed, i.e. whether or not there is occurrence dependence in unemployment. As demonstrated in the discussion surrounding equation (16), a simple test of this hypothesis can be developed by analysing the frequency distribution of waiting times to unemployment in an interval of K periods of time spent in employment for individuals who experience a single unemployment spell in the interval. If there is no occurrence dependence, duration dependence or lagged duration dependence, this distribution is uniform.

To perform this test we construct monthly employment histories. In each month, an individual is either employed, unemployed or out of the labour force.²⁴ In our sample of young men, most individuals spent little time out of the labour force so that for all practical purposes only the first two states are relevant. For each event, we can define an “occupation time”, i.e. a time spent in the state.

For the sake of concreteness, consider time spent in the employed state. Consider the first six months of calendar time. Suppose that in the first month the worker is employed, in the second and third months he is unemployed, in months four and five he is employed, and in the sixth month he is unemployed. Letting e denote employment and u unemployment, the individual’s labour market history can be represented as in Figure 4. In occupation time in the employment state, the event histories may be represented as in Figure 5. Unemployment is treated as a point event. The first spell of unemployment comes at the end of the first month of employment occupation time. The second spell comes at the end of the third month of employment occupation time. In the first three months of employment occupation time, there are two events of unemployment.

For the full 30 months of our sample, we define event histories in terms of the date of onset of their occurrence in employment occupation time. For each consecutive six-month interval of employment occupation time ($K = 6$ months in equation (16)) and for each person we record the number of events of

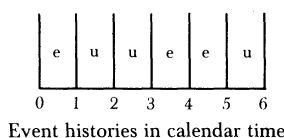


FIGURE 4. Event histories in calendar time.

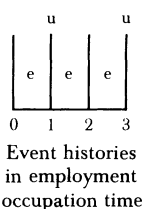


FIGURE 5. Event histories in employment occupation time.

unemployment that occur in the interval for each person.²⁵ In our sample, no person can have more than five six-month occupation time intervals, and most have fewer than five such intervals.

In a stationary environment, under the null hypothesis of no occurrence, duration or lagged duration dependence in unemployment, the conditional density of time to the first event of unemployment for those who experience one event of unemployment within the interval is a univariate uniform density that is *independent of the individual's labour market history*. We construct an aggregate distribution of time to unemployment from all six-month employment occupation time intervals *in which exactly one event occurred*. The aggregation is over people and over six-month intervals for a given person. Under the null hypothesis, the empirical distribution is uniform.²⁶

The results of this procedure are recorded in column one of Table 1. N is the number of six-month intervals used to construct the empirical distribution displayed in each column of that table. Two test statistics for a uniform distribution were used: a Kolmogorov–Smirnov test (KS) and a χ^2 test. The computed value for the KS test is 0.0543, while the critical value required to reject the null hypothesis of a uniform distribution at a 5 per cent significance level is 0.1741. Thus we do not reject the null hypothesis. The χ^2 statistic is 4.21, which is considerably less than the 5 per cent critical value of 11.07. Again, we do not reject the null hypothesis.

The remaining columns of Table 1 record the outcome of a series of tests designed to establish the robustness of our finding. The empirical results reported in column (2) are obtained by aggregating over all possible six-month employment time intervals in which one event of unemployment occurred and for which no episodes of unemployment have occurred in the preceding intervals. Column (3) reports the empirical frequency distribution of time to unemployment in a six-month interval for individuals who have had exactly one episode of

TABLE 1
FREQUENCIES OF EXIT TIMES TO UNEMPLOYMENT MEASURED IN
EMPLOYMENT OCCUPATION TIME

Month	(1)	(2)	(3)	(4)
1	12	4	7	9
2	5	3	1	5
3	13	2	9	12
4	12	7	3	10
5	10	6	3	6
6	9	5	2	8
N^*	61	27	25	50
KS	0.054	0.167	0.180	0.053
χ^2	4.21	3.89	11.71†	4.00

Definition of samples:

- (1) Frequency aggregated over all individuals and all time periods.
- (2) Frequency conditional on having zero spells up to base month.
- (3) Frequency conditional on having one spell up to base month.
- (4) Frequency aggregated over all individuals and all time periods for non-recall unemployment.

* N = number of exit times in the sample.

† Significant at 5 per cent level.

unemployment at some time prior to the interval. Our initial finding of uniformity holds up in these different sub-samples. In all cases, the KS statistic is too small to reject the null hypothesis of uniformity at the 5 per cent level of significance. Moreover, the χ^2 statistic leads to the same inference in all cases except that reported in column (3), where the null hypothesis is (barely) rejected at the 5 per cent level. Since the KS test has greater power, we may conclude that there is little evidence of non-uniform exit times from employment in our sample.²⁷

Our results are unaffected when we restrict our sample to unemployment spells that did not lead to a recall by the same employer. As is clear from column (4), the distribution of time to a non-recall unemployment spell produces test statistics that do not reject uniformity.

These results suggest that, for young men, previous unemployment experience does not affect the probability of exiting employment into the state of unemployment. In light of the fact that the economic environment was not stationary during the time period in which our data are generated and, in fact, experienced a rise in aggregate unemployment, and in light of our previous discussion of the sensitivity of the Bates–Neyman test to such non-stationarity, it is somewhat surprising that we cannot reject the null hypothesis of uniform exit times.

In principle, we can also test the hypothesis that unemployment experience does not affect the probability of exiting unemployment into the state of employment. To test this hypothesis would require repeating the type of analysis just performed for exit times measured in terms of unemployment occupation time. In practice, unemployment intervals were quite short. In our sample there are few observations that have exactly one spell of employment in unemployment time intervals of reasonable length. Although, in principle, one could use equation (16) for unemployment occupation time intervals with two or more events, we are not aware of any distribution theory for the multivariate Kolmogorov–Smirnov or χ^2 statistics that would be required for such a test. For this reason, we do not report any test of this hypothesis.²⁸

(b) Regression results

This sub-section summarizes the main empirical results of an analysis of the NLS data conducted with the regression models presented in Section II. Our main conclusions are (1) that there is no evidence of lagged duration dependence in the exit times from employment and unemployment states after corrections are made for heterogeneity; (2) that there is no evidence of occurrence dependence in the exit times from employment and unemployment states once corrections are made for sample selection bias; (3) imposing strong and intrinsically arbitrary assumptions, there is weak evidence of positive duration dependence in the exit times from employment and unemployment states. In view of our previous discussion of the fragile nature of duration dependence estimates, the final conclusion must be interpreted with some care. Moreover, owing to the limited size of our sample, both in terms of the number of individuals sampled, and the limited time period in which each individual is followed, all of our findings must be considered very tentative. The lack of statistically significant structural relationships reported here may well be a consequence of limited sample size.

We adopt the following strategy in reporting our regression results. We first report tests for mean occurrence dependence based on the regression speci-

fication given in equation (23). This specification ignores lagged duration dependence, but provides a framework for estimating duration dependence effects. We then briefly summarize the results of tests for lagged duration dependence based on equation (20a).

Tests for mean occurrence dependence in employment spells and an estimate of duration dependence. The first column of Table 2 presents empirical results from fitting regression (23) on a pooled sample of differences in lengths of *adjacent completed* employment spells. The pooling is over people and over spells for the same person. If the one-factor specification for the disturbance discussed below equation (12) is assumed, differencing eliminates the permanent component. If the permanent component contributes the dominant share of error variance, differencing tends to eliminate the selection bias, discussed in Section II(c), that arises from using only those spells with completed durations in the regression analysis.

Within the framework of equation (23), a test of the hypothesis of no mean occurrence dependence is a test of the hypothesis that variables that do not change across spells have statistically insignificant regression coefficients.²⁹ We find that the spell-invariant variables (the first five explanatory variables in Table 2) all have associated regression coefficients that are not statistically significantly different from zero. An F test for the joint hypothesis does not reject the null hypothesis. In regression results not reported here, the same findings emerge when we restrict the sample to include only individuals who are never enrolled in school in the sample, and to spells of employment that were not terminated by a spell of unemployment on recall. The same conclusions emerge if we confine our analysis to differences in lengths of employment between the first two completed employment spells (see column (2)). Column (3) reports empirical results from an attempt to correct for the selection bias that arises from restricting our sample to observations with two completed spells of employment. The procedure follows Heckman (1979).³⁰ This correction for selection bias does not alter our previous conclusions. We find no evidence that previous occurrences of unemployment affect the probability of *becoming* unemployed.

Following the procedure presented in the discussion surrounding equations (21) and (22), the estimated duration dependence parameter for spells of employment (γ_{12}) estimated from a regression fit on the first two completed spells of employment (regression coefficients reported in column (2) of Table 2) is 1.76.³¹ Taken at face value, this finding indicates that the conditional probability of becoming unemployed *increases* with the length of time spent in the employed state, and appears to contradict previous work on the unemployment-reducing effect of “firm-specific capital”.

As noted in Section II(b), spell-independent errors in measuring employment intervals tend to bias estimates of γ_{12} downward, inducing negative duration dependence in the estimates so that on average the true value of γ_{12} exceeds 1.76. However, we are inclined to be cautious about making very much of this empirical result, for two reasons. First, our estimate of the duration dependence parameter is crucially dependent on the maintained hypothesis of a Weibull model, and we have no good reason for selecting this distribution over alternative models. Second, we have not derived the sampling distribution for our estimator so that we cannot formally test the hypothesis that γ_{12} exceeds unity.

TABLE 2
REGRESSIONS ON DIFFERENCES IN LENGTHS OF EMPLOYMENT SPELLS
(*t*-statistics in parentheses)

Variable*	Pooled sample: coefficient (1)	Comparison of 1st and 2nd spell: coefficient (2)	Column (2) results corrected for selection bias: coefficient (3)
Constant	0.9989 (0.42)	1.1058 (0.38)	-2.131 (-0.32)
MSP	0.4158 (0.38)	0.5792 (0.48)	-4.049 (-0.45)
SMSA	-0.8124 (-0.21)	-0.0452 (-0.09)	-1.08 (0.50)
AGE	-0.0984 (-0.76)	-0.1343 (-0.87)	-0.444 (-0.50)
UN69	0.1567 (1.12)	0.2407 (1.32)	0.050 (0.76)
CMSP	-0.6699 (-0.70)	-0.4123 (-0.39)	-0.893 (-0.49)
CSMSA	-0.8575 (-1.25)	-0.8761 (-0.98)	2.091 (0.29)
DUN	-0.0789 (-0.53)	-0.0908 (-0.52)	-0.0294 (-0.79)
Selection variable (λ)	...	...	7.467 (0.43)
R^2	0.188	0.211	0.215
N	47	33	33

* *Key to variables:* MSP = 1 if married spouse present as of the time of the first spell; SMSA = 1 if lived in SMSA during the first spell; AGE = age as of 1969; UN69 = local unemployment rate in 1969; CMSP = 1 if marital status changed across spells; CSMSA = 1 if SMSA status changed across spells; DUN = difference in 1971 and 1969 local unemployment rates.
The first five variables—Constant, MSP, SMSA, AGE and UN69—are spell-invariant; CMSP, CSMSA and DUN change across spells.

Tests of mean occurrence dependence in unemployment spells and an estimate of duration dependence. We apply the procedures just described to differences in lengths of successive durations of unemployment. The empirical results are displayed in Table 3. The same set of explanatory variables are used as were used in the analysis of durations of employment.

In the pooled sample (results reported in column (1)), we appear to find strong evidence of mean occurrence dependence. The intercept and the coefficient on age are statistically significantly different from zero. From these results we would conclude that later spells of unemployment are longer than earlier spells, and that the coefficient on age at high school graduation declines (in absolute value) in successive spells of unemployment.

In our judgment, this finding is largely a consequence of sample selection bias. When we confine the analysis to the difference in length of the first two unemployment spells, which includes a more representative sample of the population, and which is therefore less subject to sample selection bias (see column (2)), we find no evidence of mean occurrence dependence.³² These

TABLE 3
REGRESSIONS ON DIFFERENCES IN LENGTHS OF UNEMPLOYMENT SPELLS
(*t*-statistics in parentheses)

Variable*	Pooled sample: coefficient (1)	Comparison of 1st and 2nd spell: coefficient (2)	Column (2) resulted corrected for selection bias: coefficient (3)
Constant	6.2922 (1.98)	4.4917 (1.18)	−1.059 (−0.19)
MSP	0.1974 (0.37)	0.1585 (0.28)	0.151 (0.25)
SMSA	0.1026 (0.33)	0.0666 (0.18)	0.89 (1.46)
AGE	−0.3505 (−2.09)	−0.2627 (−1.28)	0.21 (0.5)
UN69	0.0081 (0.07)	0.0581 (0.44)	0.0026 (1.74)
CMSP	0.8065 (0.70)	... **	... **
CSMSA	−1.7740 (−2.30)	−1.7802 (−2.16)	−2.404 (−2.15)
DUN	−0.0799 (−0.70)	−0.1099 (−0.86)	0.006 (−0.36)
Selection variable (λ)	...	...	3.238 (1.25)
R^2	0.194	0.194	0.21
N	50	36	36

* See Table 2 for key to variables. The first five variables—Constant, MSP, SMSA, AGE and UN69—are spell-invariant; CMSP, CSMSA and DUN change across spells.
** In our sample CMSP does not change between the first and second spells.

suspicious are further confirmed when we correct for selection bias (column (3)). We conclude that there is no evidence of mean occurrence dependence in unemployment durations.

Our estimate of duration dependence ($\gamma_{21} = 1.36$), based on the error variance from the regression reported in column (2) of Table 3,³³ suggests that the conditional probability of becoming employed increases with the length of time unemployed. This finding is consistent with a declining reservation wage for the unemployed (Kiefer and Neumann, 1979). However, for reasons already stated in our discussion of our estimate of γ_{12} , we view this result with some suspicion.

Tests for lagged duration dependence. Least-squares estimates of equation (20a) or (20b) indicate that durations of lagged employment and unemployment spells determine the length of current employment and unemployment spells. However, these conclusions vanish when the equations are re-estimated using instrumental variables for the lagged durations to correct for heterogeneity bias.³⁴ For the sake of brevity we do not report these results in this paper. From these results we conclude that there is no evidence of true lagged duration dependence in our sample.

(c) *Qualifications of empirical results*

We warn against any generalization of these results to explain life-cycle labour market phenomena. Our empirical results may be due, in part, to the fact that we study a sample of very young men who are just beginning their labour market experience. It is quite possible that as these men age and accumulate different types of labour market information state dependence may be more in evidence.

More importantly, the samples on which we have conducted our empirical analyses are small. The small size of the sample is largely a consequence of the fact that the designers of the National Longitudinal Study of Young Men did not collect monthly labour market histories for more than a handful of young men in the period after they graduated from high school. It must be acknowledged that the small size of the sample makes it very hard to detect significant statistical relationships, and makes it more likely to find the sort of randomness in the data that we report in this paper. Investigation of larger data sets with longer labour market histories would provide more convincing evidence on the existence of structural state dependence.

SUMMARY

This paper presents a continuous-time model of heterogeneity and state dependence that builds upon, and extends, previous work on continuous-time discrete-state models. In order to determine how previous labour market experience affects current labour market behaviour, we construct a continuous-time model of labour market dynamics that admits more general types of state dependence than are admitted in Markov models. Three types of state dependence are considered in addition to standard Markov dependence: (a) *occurrence dependence*—defined as the effect of previous occupancy of labour force states on the probability of transition from a given state; (b) *duration dependence*—defined as the effect of current duration in a state on the probability of exit from the state; and (c) *lagged duration dependence*—defined as the effect of the durations in previous states on the probability of transition from a current state.

Different conditions are required to identify the types of state dependence considered in this paper. Tests for occurrence dependence can be conducted under quite general conditions. Tests for lagged duration dependence require satisfaction of identification criteria formally equivalent to the criteria used to discriminate between serial correlation and distributed lag models in time series econometrics (Griliches, 1967). Tests for duration dependence or “specific capital” require making strong, and intrinsically arbitrary, assumptions about distribution functions for population heterogeneity and “true state dependence”.

We applied our procedures to analyse continuous work histories taken from a sample of young men who just graduated from high school. For this sample, we find no evidence that previous occurrences of unemployment or their duration affect future labour market behaviour once we control for sample selection bias and heterogeneity bias.

All of the analysis in this paper has been conducted for a two-state model. The framework proposed herein can be generalized to a multiple-state model. This generalization is reported elsewhere (Heckman and Borjas, 1979; Flinn and

Heckman, 1980). These papers also consider several different computationally efficient maximum likelihood estimates for the two-state and multiple-state models.

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NOTES

¹ The first condition is equivalent to the condition that in a "unit of time" an employed worker either remains employed (with probability $a_{11} + 1$) or becomes unemployed (a_{12}) so that $(a_{11} + 1) + a_{12} = 1$. A parallel argument can be made about the second condition so that $(a_{22} + 1) + a_{21} = 1$.

² Usually these characteristics are assumed to be gamma-distributed.

³ The intuitive appeal is based on the idea that transactions costs inhibit rapid transitions over short intervals. If $a_{12} \rightarrow \infty$, and $a_{21} \rightarrow \infty$, one can generate a Bernoulli model from the continuous-time Markov set-up.

⁴ "Cause" is a loaded word. As noted by Hume (1754), a timing relationship between events, of the sort described in the text, or the sort used in "causality tests" in time-series analysis, has no logical basis. Thus our use of the word is conventional, but misleading.

⁵ This is an artefact of the two-state model used in this paper and is not a general result (see Heckman and Borjas, 1979 or Flinn and Heckman, 1980, for a multiple-state model).

⁶ Note that in a model with more than two states, the general semi-Markov model allows for an exit time distribution that depends on the origin state and the destination state. In a two-state model, this distinction is irrelevant.

⁷ Silcock (1954) demonstrates this bias for a model of duration dependence. For more discussion of this bias, see Heckman and Borjas (1979) and Flinn and Heckman (1980).

⁸ Actually, the content of $Z_{ij}^{(l)}$ and $V_{ij}^{(l)}$ may depend on both the origin state i and the destination state j . In the two-state model developed in this paper, this distinction is irrelevant.

⁹ Consider the following example. Let

$$F(t|\theta) = 1 - \int_{t/\sqrt{\theta}}^{\infty} \frac{2}{\sqrt{2\pi}} \exp(-z^2/2) dz, \quad t \geq 0.$$

The structural hazard rate for the model is clearly not independent of t . Let the θ parameter be exponentially distributed $p \exp(-p\theta)$. Then

$$G(t) = \int_0^{\infty} F(t|\theta) p e^{-p\theta} d\theta = 1 - \exp(-tp^{1/2})$$

so that the empirical hazard function reveals no duration dependence while the structural hazard function generates true duration dependence. Many other such examples can be constructed, so that this example is not unique.

¹⁰ Thus if we start with the arbitrary initial condition $\ln t_{ij}^{(l)} = 0$,

$$\ln t_{ij}^{(l)} = \sum_{k=0}^{l-1} \beta^k \varepsilon_{ij}^{(l-k)}$$

so that letting $E(\varepsilon_{ij}^{(l)})^2 = \sigma$, $E(\ln t_{ij}^{(l)}) = 0$ and

$$E(\ln t_{ij}^{(l)})^2 = \sigma \sum_{k=0}^{l-1} \beta^{2k}$$

so there is no mean occurrence dependence but there is occurrence dependence. However, if we assume that $\ln t_{ij}^{(0)}$ has mean zero and variance $\sigma/(1 - \beta^2)$, $E(\ln t_{ij}^{(0)}) = 0$ $E\{(\ln t_{ij}^{(0)})^2\} = \sigma/(1 - \beta^2)$ so there is no occurrence dependence but there is still lagged duration dependence.

¹¹ By “nature” we mean some estimable parametric relationships that permit us to introduce exogenous variables to control for the non-stationarity.

¹² One could argue that the changes in the exogenous variables between the m th and $(m + 1)$ st employment spells are due to the fact that an individual experiences a spell of unemployment. Thus the changes in the exogenous variables between spells can be attributed to the m th unemployment episode, so that evidence against uniformity due to explanation (b) is consistent with occurrence dependence. This argument is more a comment on the specification of equation (18) than an argument in support of changes in exogenous variables as a source of occurrence dependence. It is, indeed, unlikely that truly exogenous variables would change across spells in the way required to write down equation (18). More likely, exogenous variables would change in calendar time and would not be constant within spells. A more general model for time-varying variables of the sort considered in Section I(c) would be appropriate, and if this more general model is the correct model, equation (18) is mis-specified. However, if the exogenous variables are slowly changing, the mis-specification may not be severe if $\mathbf{Z}^{(m)}$ is defined as the value of an underlying time-varying exogenous variables at the onset of spell m (or if some other convention is adopted). In this case, in which the exogenous variables change in a way independent of the spells experienced, one would not want to attribute changes in $\mathbf{Z}$ to the occurrence of a spell of unemployment. This discussion underscores the point that the Bates–Neyman procedure is not robust with respect to changing values of exogenous variables. Note, however, that if we augment the exogenous variables to include lagged durations, rejection of uniformity from this source *would* be evidence in support of structural state dependence.

¹³ The discussion in the preceding note is germane.

¹⁴ The proof of this assertion is straightforward and hence is omitted. Constellations of occurrence, lagged duration and duration dependence that lead to a uniform distribution occur on a set of measure zero, and hence can be ignored.

¹⁵ B_1 is $\Gamma'(1)$, the derivative of the gamma function evaluated at unity ($=0.577\dots$). B_2 is $(d\ln \Gamma(\theta))/d\theta$ evaluated at $\theta = 1$ ($=\pi^2/6$). Equation (20a) can be verified using standard Laplace transform methods to derive the conditional mean. The distribution of $W_{ij}^{(0)}$ is derived in Heckman and Borjas (1979) or Flinn and Heckman (1980).

¹⁶ We note, parenthetically, that because of the non-normality of the error, the least-squares estimators of the slope coefficients are not efficient. In fact, in his comments on this paper, Amemiya has proved that, if $\gamma_{ij}^{(0)}$ is known, the asymptotic relative efficiency of least squares is $6/\pi^2$ for a model without heterogeneity. See Heckman and Borjas (1979) or Flinn and Heckman (1980).

¹⁷ An alternative procedure is to impose the requirement that the intercept term is zero in the $\beta_{ij}^{(0)}$ parameter vector. Then $\gamma_{ij}^{(0)}$ can be estimated since B_1 is known.

¹⁸ The proof is straightforward. Let S_{ll} be the estimated residual variance from the regression for the l th duration interval, and let $S_{ll'}$ be the estimated covariance between the residuals of the l th and l' th duration regressions. Define the inter-equation residual correlation as

$$r_{ll'} = \frac{S_{ll'}}{(S_{ll} S_{l'l'})^{1/2}}.$$

In this notation, the estimators of the parameters C_{ij}^2 , σ_ϕ^2 , $\gamma_{ij}^{(0)}$ and $\gamma_{ij}^{(l)}$ are (letting the caret denote estimate)

$$\begin{aligned} \widehat{C_{ij}^2 \sigma_\phi^2} &= \frac{B_2 r_{ll'}}{1 - r_{ll'}} \\ \hat{\gamma}_{ij}^{(0)} &= \left\{ \left(\frac{B_2}{S_{ll}} \right) \left(\frac{1}{1 - r_{ll'}} \right) \right\}^{1/2} \\ \hat{\gamma}_{ij}^{(l)} &= \left\{ \left(\frac{B_2}{S_{l'l'}} \right) \left(\frac{1}{1 - r_{ll'}} \right) \right\}^{1/2} \end{aligned}$$

where positive values of square roots are to be used everywhere in evaluating these expressions to ensure that estimated values of $\hat{\gamma}_{ij}^{(0)}$ and $\hat{\gamma}_{ij}^{(l)}$ are positive, a requirement of the Weibull model. Note further that negative $r_{ll'}$ is evidence against the Weibull model with the factor structure stated in the text.

¹⁹ Clearly, if data are available on unemployment spells, and the heterogeneity component in the unemployment equation is correlated with ϕ , log duration in unemployment is also a valid instrument.

²⁰ To see this, suppose that the disturbance term of the log-normal regression analogue of equation (21) is $F + N_{ij}^{(l)}$ where F is a permanent component with mean zero and variance σ_F^2 , and the first two moments of $N_{ij}^{(l)}$ are presented in equations (20b). Applying the procedure discussed in note 18,

$$\begin{aligned}\text{plim } \widehat{C_{ij}^2 \sigma_\phi^2} &= \frac{B_2 \sigma_F^2}{(\sigma_F^2 + \tilde{\sigma}_{ij}^{(l)})^{1/2} (\sigma_F^2 + \tilde{\sigma}_{ij}^{(l)})^{1/2} - \sigma_F^2} \\ \text{plim } \hat{\gamma}_{ij}^{(l)} &= \left[\frac{B_2}{(\sigma_F^2 + \tilde{\sigma}_{ij}^{(l)})^{1/2}} \frac{(\sigma_F^2 + \tilde{\sigma}_{ij}^{(l)})^{1/2}}{(\sigma_F^2 + \tilde{\sigma}_{ij}^{(l)})^{1/2} (\sigma_F^2 + \tilde{\sigma}_{ij}^{(l)})^{1/2} - \sigma_F^2} \right]^{1/2} \\ \text{plim } \hat{\gamma}_{ij}^{(l')} &= \left[\frac{B_2}{(\sigma_F^2 + \tilde{\sigma}_{ij}^{(l')})^{1/2}} \frac{(\sigma_F^2 + \tilde{\sigma}_{ij}^{(l')})^{1/2}}{(\sigma_F^2 + \tilde{\sigma}_{ij}^{(l')})^{1/2} (\sigma_F^2 + \tilde{\sigma}_{ij}^{(l')})^{1/2} - \sigma_F^2} \right]^{1/2}\end{aligned}$$

The estimates do not correspond to parameters of economic interest. However, if $\hat{\gamma}_{ij}^{(l)} = \gamma_{ij}^{(l')}$, we have evidence that innovation variances are stationary across spells.

²¹ One requires one such independent variable for each lagged spell included in the regression, and no collinearity among the instruments.

²² Obviously, one must suppress the coefficients associated with variables that do not change between spell l and l' in the first set of regressors. Note further that if the model in the text is augmented to include lagged durations as regressors, one will, in general, need to instrument the lagged durations to achieve consistent parameter estimators.

²³ See US Department of Labor (1970) for a detailed description of the data.

²⁴ Clearly, the use of monthly data may involve time aggregation bias. In principle, a person could be in all three states in a given month. We use a procedure of allocating the full month to the state in which the individual spends most of his time in the month.

²⁵ The empirical results reported below are not sensitive to the choice of $K = 6$: we obtain the same findings for $K = 4$ or $K = 10$.

²⁶ Recall that, under the assumptions, the conditional distributions do not depend on individual heterogeneity components.

²⁷ See Kendall and Stuart (1973, p. 476). Another advantage of the KS test is that it is an exact small sample test while the χ^2 test is a large sample test. Given the size of our samples, this is a relevant consideration that favours tests based on the KS statistic.

²⁸ An "eyeball test" suggests that we would not, in fact, reject the hypothesis stated in the first sentence of this paragraph.

²⁹ The inference from this test is not affected when the equations are re-estimated by GLS, correcting for the serial correlation induced by the differencing procedure.

³⁰ Assuming a normally distributed population regression disturbance, and assuming that the probability of having two completed employment spells can be well approximated by a Probit probability which is a function of all the exogenous variables defined below the table, we estimate the conditional mean of the regression disturbance in our selected sample, and enter it as a regressor in our model.

³¹ The estimate is derived from the following considerations. Assuming no occurrence dependence, and a one-factor model, the variance of the disturbance term in the *differenced* duration regression is $(2) (\pi^2/6) (1/\gamma_{12}^2)$, since differencing eliminates the permanent component, and the innovation variance is $(\pi^2/6) (1/\gamma_{12}^2)$. The estimated error variance is 1.778. Hence $\hat{\gamma}_{12} = 1.76$. Note that independently distributed components of measurement error in durations bias estimates of $\hat{\gamma}_{12}$ downward by inflating the size of the disturbance variance.

³² F tests do not reject the hypothesis of no mean occurrence dependence.

³³ The estimated error variance is 1.065. For the procedure used to derive this estimate see note 31.

³⁴ This result is robust for a variety of choices of instruments, and holds up when corrections are made for sample selection bias. A summary of these results is available from the authors.

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